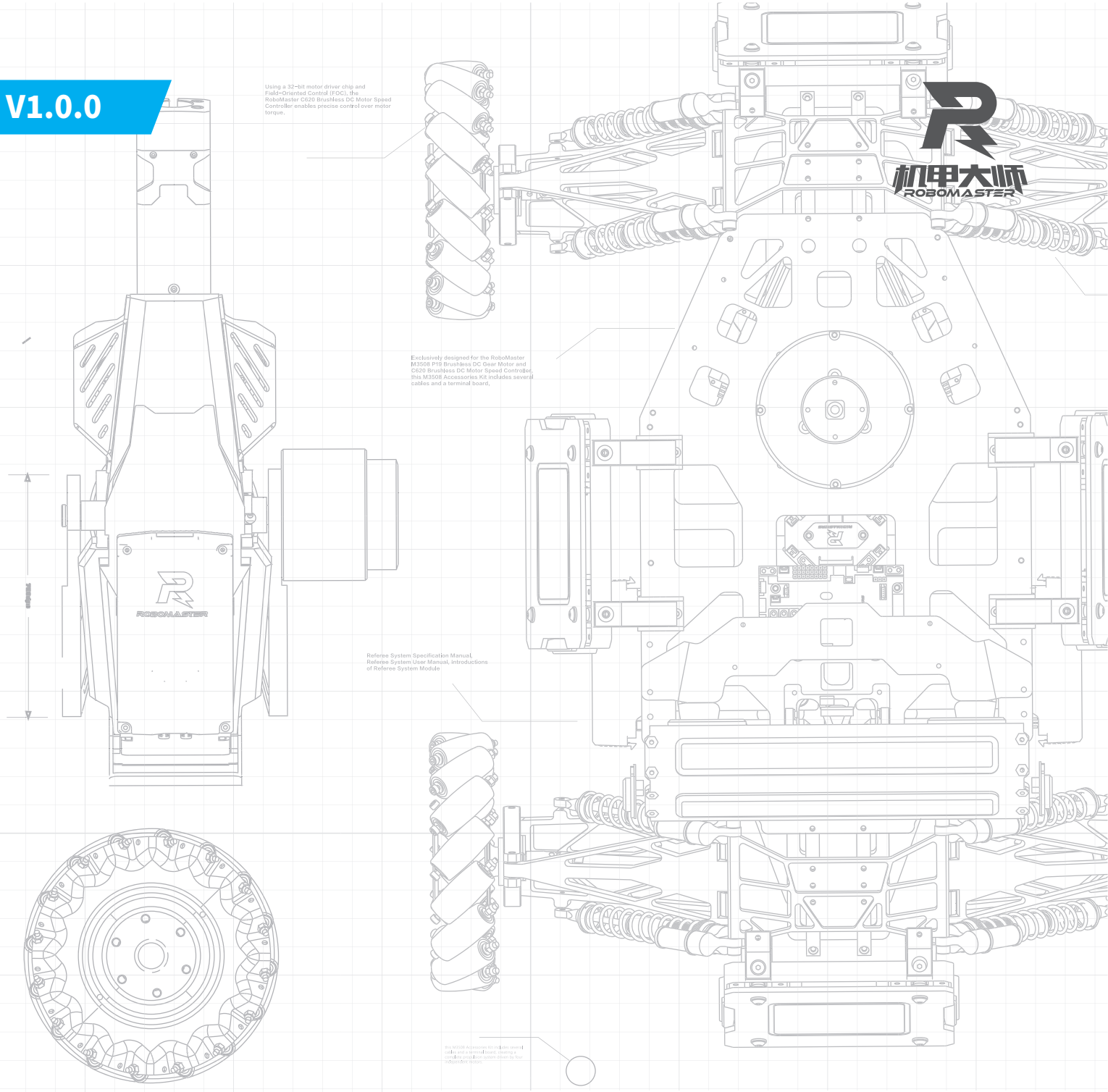
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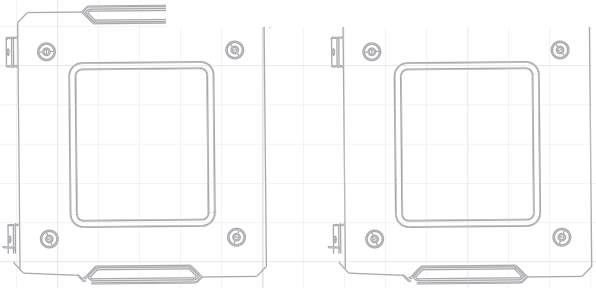


26th National University Student Robot Competition

ROBOMASTER 2027

RoboMaster University League

Competition Rules Manual



Compiled by the RoboMaster Organizing Committee

Published October 2026

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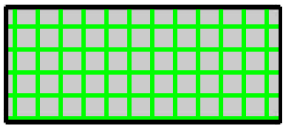
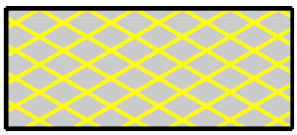
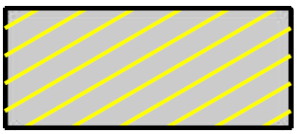
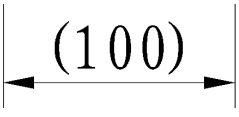
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Reading Notes

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Field Drawing Legend

		
shared Buff Point	one-side Buff Point	Shared Restricted Zone
		
One-side Restricted Zone	The horizontal plane where the ground of Battlefield is located is the lowest plane of Battlefield	Dimensions are for reference only

Revision Log

Date	Version	Revision Record
2026.10.15	V1.0.0	First release

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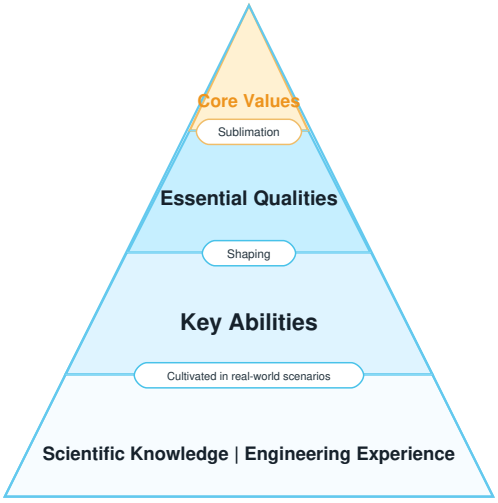
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1 Introduction

1.1 About the Competition

The National College Student Robot Competition RoboMaster University League (RMUL, RoboMaster University League) is initiated by DJI and hosted by local academic institutions and universities, drawing in surrounding universities to participate. It aims to promote regional exchange of university robotics technology, foster a strong academic atmosphere, and support regional scientific and technological innovation.

RMUL is committed to cultivating outstanding young engineers and encourages participating team members to combine the knowledge they have learned with practical applications and achieve continuous growth during the competition.



- **Core Values**
  Investigate things to extend knowledge, strive for excellence, reflect and examine oneself
  - Develop the habit of deep thinking and cultivate the ability to identify the root cause of problems
  - Pursue excellence driven by high goals and maintain an active, positive mindset
  - Face oneself honestly and reflect on the limits of one's own abilities
- **Essential Qualities**
  Proactive, positive, seeking truth from facts, keeping an open mind
- **Key Abilities**
  Thinking, communication and expression, problem solving, autonomous learning, forward design

Figure 1-1 Talent Development Philosophy

The RMUL 2027 season has two challenge events: the 3V3 Match and the Infantry Match.

- In the 3V3 competition, both sides need to independently develop Heavy Robot, Infantry Robot and Sentry Robot, and conduct tactical confrontation by controlling the robots to launch projectiles to attack the opponent's robots in the designated Battlefield. At the end of the game, the team that controls Battlefield wins the game.
- In the Infantry Match, each side must independently develop an Infantry Robot and control it to launch projectiles in a 1V1 confrontation on the designated Battlefield. When the match ends, the side whose Infantry Robot has more remaining HP wins.

1.2 Main Changes in the New Season

Compared with RMUL 2026, the main adjustments of RMUL 2027 are as follows:

3V3 Match

- Added Heavy Robot, canceled Hero Robot, and merged the original custom controller and custom client into Custom Terminal;
- Added new tunnels and workbench, adjusted the overall layout of Operator Room;
- Added the Heavy Robot special mechanism and Field Buffs mechanism, and adjusted the control zone and Control Points mechanisms.

Infantry Match

Added HP Recovery mechanism

Cancellation of the Engineer Challenge

1.3 About the Competition Rules Documents

Competition rules documents include Rules Manual, Participant Manual, Robot Building Specifications Manual, communication protocols, and supplements to the above documents. Competition rules documents apply to all participating teams, referees, event staff and other partners.

During non-competition periods, RMOC will update competition rules documents from time to time based on the actual situation.

RMOC has the final right to interpret competition rules documents. During the competition, only the Chief Referee will answer competition rules documents on behalf of RMOC. Any questions about competition rules documents can only be consulted by the Chief Referee.



For more reference materials about RMU, please refer to the "Rules/Information Center". This site is built and maintained by RoboMasterRMOC. It centrally provides official information required for competition and is designed to help teams obtain information efficiently. Content coverage:

- Competition rules: Rules Manual, Robot Building Specifications Manual, Participant Manual, communication protocol.
- Competition products: Referee System (including onboard modules and match engine), robot kits and accessories, RM Assistant product details, data downloads, purchase and after-sales policies.
- Technical Assessment: Technical Assessment details and instructions for each link.
- Brand promotion: Brand promotion material package.

- Investment Guide: Investment Manual, Investment Guide.
- Training system: competition strategies, course salons, and open source materials.

1.4 About Q&A

When the participating team and other relevant personnel have questions about competition rules documents, they can raise questions through official channels, and RMOC will answer the questions.
The specific process is as follows:

1. The questioner fills out the questionnaire and submits questions:

   https://qingflow.com/f/fpckalpss002?exWsld=1SdM5_PB-NwUtnyPn2b8NA

2. The Organizing Committee will post its responses at:

   https://qingflow.com/appView/fpckalpss002/shareView/fpckam8gs001

Timeliness statement:

- For questions submitted during the competition period, RMOC will provide feedback within 1 calendar day;
- For questions submitted during the preparation period, RMOC will provide feedback within 5 working days.



- Competition period: starts from the registration day of the event and ends on the last competition day of the event.
- Preparation period: the period outside the competition period.

Note: The competition period and preparation period for different events and stages are calculated separately.

Questions and answers have the same effect as competition rules documents. When there are differences between the Q&A and competition rules documents, the latest published Q&A or competition rules documents shall prevail. Each season's Q&A applies only to that season.

2 Important Concepts

This section briefly explains important common concepts related to the competition rules. For details, use the keywords to find the relevant chapters.

Table 2-1 Overview of Important Concepts

Concept	Definition
Robot	
Robot	Collective term for the Ground Robots and Auxiliary Equipment that participating teams bring and use in the competition.
Ground Robot	A robot that operates independently within the competition field, including the Heavy Robot, Infantry Robot and Sentry Robot.
Auxiliary Equipment	Equipment that assists the operation of Ground Robots and does not directly take part in the competition confrontation, including: the Custom Terminal.
Robot main states	- Alive : the state in which the robot's Referee System Main Controller Module is normally connected to the Referee System server and its HP is not 0. • Defeated : the state in which the robot's HP is 0 (causes include: Armor Module being attacked, being hit by collision, Referee System onboard module going offline, Ejection). • Ejection : the state in which the robot has received a Red Card Warning. • Temporary activation : the state in which the robot's chassis and gimbal are temporarily powered on after the robot is Defeated. At this time the robot's launching mechanism is in the power-off state. • Abnormal offline : the state in which, during the match, the Referee System Main Controller Module cannot connect to the Referee System server because the robot lost power or for other reasons. Note: After a robot is Defeated, the Referee System cuts off the power output to the robot (except the Mini PC).
Referee System	The Referee System is an electronic judging system for robot competitions that integrates computing, communication and control. It includes the onboard modules and the competition engine installed on PC hosts (server, referee client, player client), and has functions such as monitoring robot power, projectile launches and damage, and automatically determining the winner according to the competition rules.
Multi-robot communication	The mode of interaction in which robots communicate with each other through the Referee System serial port.
Chassis	The mechanism that carries the robot's power system and its accessory parts and supports the robot body.

Concept	Definition
Chassis Power	The power consumed by a mechanism that uses rotary motion and ground friction to move or rotate a robot horizontally on a plane. See the Chassis Power definition in the 'Referee System Installation Specifications' section of the RoboMaster 2027 University Series Robot Building Specifications Manual.
Launching mechanism	A mechanism that enables a projectile to leave the robot along a fixed path at a certain initial speed.
Launching mechanism locked	If the launching mechanism is locked, it is powered off.
Launching mechanism unlocked	If the launching mechanism is unlocked, power on or off is determined based on its Projectile Allowance status.
Launch	The act of a robot firing a projectile. In the later 'Competition Mechanisms' section, 'Launch' means a firing action detected by the Referee System.
Initial Launch Speed	The speed value detected by the relevant Referee System module after the projectile has finished accelerating.
Barrel Heat	Barrel Heat increases as the robot fires projectiles and is used to limit continuous firing.
Projectile Allowance	The number of projectiles that each robot can currently launch.
Initial HP	The HP value assigned to a robot by the Referee System at the start of a match. Note: The Referee System rounds HP to the nearest integer when settling it.
Current HP	The real-time HP value of the robot.
Max HP	The maximum value to which the robot's HP can be restored.
Experience	The value a robot must accumulate to level up; it can be obtained in multiple ways.
Invincible	The robot is immune to damage from projectile attacks and collisions.
Attack	The act of a robot firing a projectile.
Destroy	One robot attacks the other robot's Armor Module, causing its HP to be reduced to 0.
Activation	An Alive robot reaches a Buff Point, and its Field Interaction Module detects a Field Interaction Module card within the zone or other activation conditions are met. meet other conditions.

Concept	Definition
Rigid connection	A mechanical connection formed between robots during a match, such that when one robot moves in any direction it remains mechanically connected to the other robot.
Ramming	An active collision by one side's robot during the match.

Battlefield

Buff Point	A zone that produces certain special effects after being activated by one side's robot during the match.
Restricted Zone	A zone that robots are prohibited from entering.
Field Props	Elements that make up the Battlefield.

personnel

Arbitration Committee	The Chief Referee and other relevant RMOC personnel, responsible for handling matters such as Appeals.
Referee	Personnel who maintain order on the field and enforce the competition rules.
Chief Referee	During the competition period, the person with final authority to interpret the competition rules documents.
Head Referee	The primary referee responsible for maintaining order on the field and enforcing the competition rules.
Chief Inspector	The referee responsible for leading and assigning Pre-Match Inspection work, who has final authority to interpret the inspection standards.
participating team	Teams that registered for the current season and were entered into the registration system.
Participant	People who registered for the current season and were entered into the registration system.
Battlefield personnel	Official Team Members, Team Mentors and teachers who registered for the current season and were entered into the registration system, and who may enter the waiting area and Battlefield. Note: Member roles differ by event; refer to the RoboMaster 2027 University League Participant Manual.
Operator	A Battlefield person responsible for controlling robots during a match.
Offending party	The side or participating team that violates a rule.
Violating personnel	A participant who violates the rules.

Concept	Definition
Competition process	
Round	A complete match comprising the Setup Period, Referee System Self-Check Period, Five-Second Countdown and Round period.
Match	Depending on the format, a game may consist of several rounds.
Official technical pause	A technical pause initiated by the Head Referee during the Setup Period or Referee System Self-Check Period.
participating team technical suspension	A technical pause requested by a participating team during the Setup Period.
Routine Battle Damage	A failure of a Referee System onboard module during the inter-round Setup Period or Round period of a match.
Determination of victory or defeat	
attack damage	HP deducted from one side's robot by projectile attacks. Note: HP deduction on one side's robot caused by its own rule violation is counted as attack damage dealt by the opposing side.
non-attack damage	HP deduction caused by impacts on Armor Modules, Referee System modules going offline, and similar causes.
Total remaining HP	The total remaining HP of all Alive robots on one side at the end of each round.
Net remaining HP	At the end of each round, one side's remaining HP minus the opposing side's remaining HP.
Control Points	A measure of one side's total control over the Battlefield by occupying positions, calculated at the end of each round.

3 3V3 Match

During the time-limited five-minute Round, the two sides engage in tactical confrontation by controlling robots in the core competition area, the Battlefield. The side with more Battlefield Control Points causes the opposing side to lose Control Points. The side that first reduces the opponent's Control Points to zero wins.

3.1 Robots and Operators

RoboMaster emphasizes teamwork among robots and requires balanced cooperation. For robot building specifications, see the

'RoboMaster 2027 University Series Robot Building Specifications Manual'.

The robot and Operator lineups are as follows:

Table 3-1 3V3 Match Robot and Operator Lineup

Type	Number	Number	Maximum number of reserve robots quantity	Maximum number of equipped Custom Terminals	Maximum number of reserve Custom Terminals	Operator Lineup
Heavy Robot	1	1	1	1	1	1 person per robot
Infantry Robot	3	1	1	1	1	1 person per robot
Sentry Robot	7	1	1	1	1	0 persons per robot

The total number of robots that each team can register in an independent stage is: 4.

In each match, the minimum lineup in the first round is 2 robots.

Operator can only be held by Battlefield personnel other than the supervising teacher in the current participating team.

After each game, the Operator can be replaced from the current Battlefield personnel except the supervising teacher.

3.1.1 Heavy Robot

After successfully installing a modular 42mm launching mechanism, Heavy Robot is the only robot in Battlefield that can launch a 42mm projectile.

Table 3-2 Key Features of the Heavy Robot

Key feature	Description
Initial zone	Starting Zone
Mode of operation	Unrestricted; up to 1 Remote Controller and 1 Custom Terminal may be configured

<table>
<tr><th>Key Characteristics</th><th>Description</th></tr>
<tr><td>Multi-robot communication</td><td>Allowed</td></tr>
<tr><td>Launching mechanism</td><td>1 modular 42mm launching mechanism</td></tr>
<tr><td>Performance</td><td>
<table>
<tr><th>Status</th><th>Max HP</th><th>Chassis Power upper limit</th><th>Initial Launch Speed upper limit</th><th>Barrel Heat Limit</th><th>Barrel Cooling Rate (/s)</th></tr>
<tr><td>Start of match</td><td>100</td><td>50</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>After successfully installing a modular 42mm launching mechanism</td><td>450</td><td>50</td><td>16.5</td><td>250</td><td>66</td></tr>
</table>
</td></tr>
<tr><td>HP deduction and over-limit penalty mechanism</td><td>- Initial Launch Speed exceeding the limit: applicable
- Barrel Heat exceeding the limit and cooling: applicable
- Chassis Power exceeding the limit: applicable
- Attack damage or collision: applicable
- Referee System onboard module offline: applicable
- Penalty Damage: applicable
See ‘3.3.1 HP Deduction and Exceeding-the-Limit Penalty Mechanisms’ for details.</td></tr>
<tr><td>HP Recovery and Revival mechanisms</td><td>- HP Recovery method: HP can be recovered by activating the Supply Zone.
• Revival method: progress-bar Revival. See "3.3.2 HP Recovery and Revival Mechanisms".</td></tr>
<tr><td>Projectile loading and replenishment</td><td>During the three-minute Setup Period, 42mm projectiles can be preloaded; the initial Projectile Allowance is 0.</td></tr>
<tr><td>Activatable Buff Points</td><td>- Control Zone (see "3.3.4 Control Zones and Control Points Mechanism")
• Supply Zone Buff Point (see "3.3.2 HP Recovery and Revival Mechanisms")</td></tr>
</table>

Table 3-3 Key characteristics of the modular 42mm launching mechanism

Key Characteristic	Description
Initial area	Workbench
Projectile loading and supply	During the three-minute Setup Period, 42mm projectiles can be preloaded; the initial Projectile Allowance is 0.

3.1.2 Infantry Robot

The Infantry Robot can fire 17mm projectiles.

Table 3-4 Key characteristics of the Infantry Robot

Key Characteristic	Description
Initial area	Starting Zone
Operating mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal may be configured
Multi-robot communication	Allowed
Launching mechanism	1 17mm launching mechanism
Initial Launch Speed limit (m/s)	25
Performance	After the Setup Period begins, the Operator of the Infantry Robot can select the robot's chassis type (HP Priority or Power Priority) and launching mechanism type (Burst Priority or Cooling Priority). After the Round begins and the chassis and launching mechanism types have been selected, they cannot be changed for the rest of the round. If no chassis or launching mechanism type is selected, then after the Round begins the chassis type defaults to "HP Priority" and the launching mechanism type defaults to "Cooling Priority". Chassis type • Power Priority: • Initial HP / Max HP: 300 • Chassis Power limit: 90W • HP Priority: • Initial HP / Max HP: 350 • Chassis Power limit: 75W

Key Characteristic	Description
	Launching mechanism type - Burst-priority mode: - Barrel Heat Limit: 230 - Barrel Cooling Rate: 14/s - Cooling-priority mode: - Barrel Heat Limit: 88 - Barrel Cooling Rate: 24/s
HP deduction and exceeding-the-limit penalty mechanisms	- Initial Launch Speed exceeding the limit: applicable • Barrel Heat exceeding the limit and cooling: applicable • Chassis Power exceeding the limit: applicable • Attack damage or collision: applicable • Referee System onboard module offline: applicable • Penalty Damage: applicable See "3.3.1 HP Deduction and Exceeding-the-Limit Penalty Mechanisms".
HP Recovery and Revival mechanisms	- HP Recovery method: HP can be recovered by activating the Supply Zone. • Revival method: progress-bar Revival. See "3.3.2 HP Recovery and Revival Mechanisms".
Projectile loading and supply	During the three-minute Setup Period, 17mm projectiles can be preloaded; the initial Projectile Allowance is 0.
Activatable Buff Points	- Control Zone (see "3.3.4 Control Zones and Control Points Mechanism") • Supply Zone Buff Point (see "3.3.2 HP Recovery and Revival Mechanisms")

3.1.3 Sentry Robot

The Sentry Robot must operate fully automatically and can fire 17mm projectiles.

Table 3-5 Key characteristics of the Sentry Robot

Key Characteristic	Description
Initial area	Starting Zone
Operating mode	Automatic operation; at most 1 Remote Controller may be configured for debugging
Multi-robot communication	Not allowed
Launching mechanism	1 17mm launching mechanism
Initial Launch Speed limit (m/s)	25
Performance	- Initial HP / Max HP: 400 • Chassis Power limit: 100W • Barrel Heat Limit: 260 • Barrel Cooling Rate: 30/s
HP deduction and exceeding-the-limit penalty mechanisms	- Initial Launch Speed exceeding the limit: applicable • Barrel Heat exceeding the limit and cooling: applicable • Chassis Power exceeding the limit: applicable • Attack damage or collision: applicable • Referee System onboard module offline: applicable • Penalty Damage: applicable See "3.3.1 HP Deduction and Exceeding-the-Limit Penalty Mechanisms".
HP Recovery and Revival mechanisms	- HP Recovery method: HP can be recovered by activating the Supply Zone. • Revival method: progress-bar Revival. See "3.3.2 HP Recovery and Revival Mechanisms".
Projectile loading and supply	- During the three-minute Setup Period, 17mm projectiles can be preloaded.

Key Characteristic	Description
	During the five-minute Round, the initial Projectile Allowance is 750 and cannot be increased during the round.
Activatable Buff Points	- Control Zone (see "3.3.4 Control Zones and Control Points Mechanism") • Supply Zone Buff Point (see "3.3.2 HP Recovery and Revival Mechanisms")

3.1.4 Custom Terminal

The Custom Terminal (formerly the Custom Controller and Custom Client) is a multipurpose set of equipment built by participating teams. It monitors robot motion and status, controls robot actions, receives match-related information and robot camera video feeds sent by the Referee System's match engine server, and sends match-specific interaction commands to that server to expand the possibilities for human-computer interaction.

A typical Custom Terminal may consist of a display device, a computing platform (including memory, controller and processor) and a set of input devices. It may also be integrated, such as a handheld tablet with a network port.

Example: A team builds a Custom Terminal that receives robot camera video and displays it on a self-made monitor. It overlays a self-made UI displayed through an unofficial link and controls the robot with a matching controller. As a result, the team can complete the match without using the official Referee System player terminal or Remote Controller.

Custom Terminals in the Operator Room can be connected to and communicate with each other by wire. They can also be wired to the match engine server and send and receive relevant content through the RJ45 interface in the Operator Room.

A Custom Terminal can receive match information shown on the Referee System player terminal and available to its user, such as both sides' HP, the user's own Economy, remaining match time and key match events.

A Custom Terminal can send match commands supported by the player terminal to the Referee System, such as selecting a Performance System before the match or redeeming Projectile Allowance during the match.

3.2 Competition Field

3.2.1 Battlefield Overview



- Except for special markings, all dimensional errors described in the full text are within ±10%, the angular error of the Battlefield structure is within ±3°, the angular error of the Battlefield props is within ±1°, and the structural error of dimensions less than 100mm is within ±5mm. Battlefield states that the drawing size parameters are in mm. In order to ensure the stability and safety of the Battlefield, the detailed dimensions and craftsmanship of the Battlefield may be adjusted during actual production. For Battlefield elements/ dimensions that are not specified in the drawings, please refer to the actual Battlefield.
- The Battlefield has a centrally symmetric layout. Descriptions and explanatory diagrams of Battlefield modules throughout this manual use the Red Side as the example; the Blue Side is the same.
  The Battlefield environment is complex, and there may be uneven magnetic fields, venue air-conditioning winds and other non-completely symmetrical environmental factors.

The core competition area in the 3V3 Match is called the Battlefield. The Battlefield is 12 m long and 8 m wide. It has a wooden structure with 2 mm-thick PVC floor covering and metal protective layers on some structures. The area contains Supply Zones and Control Zones.

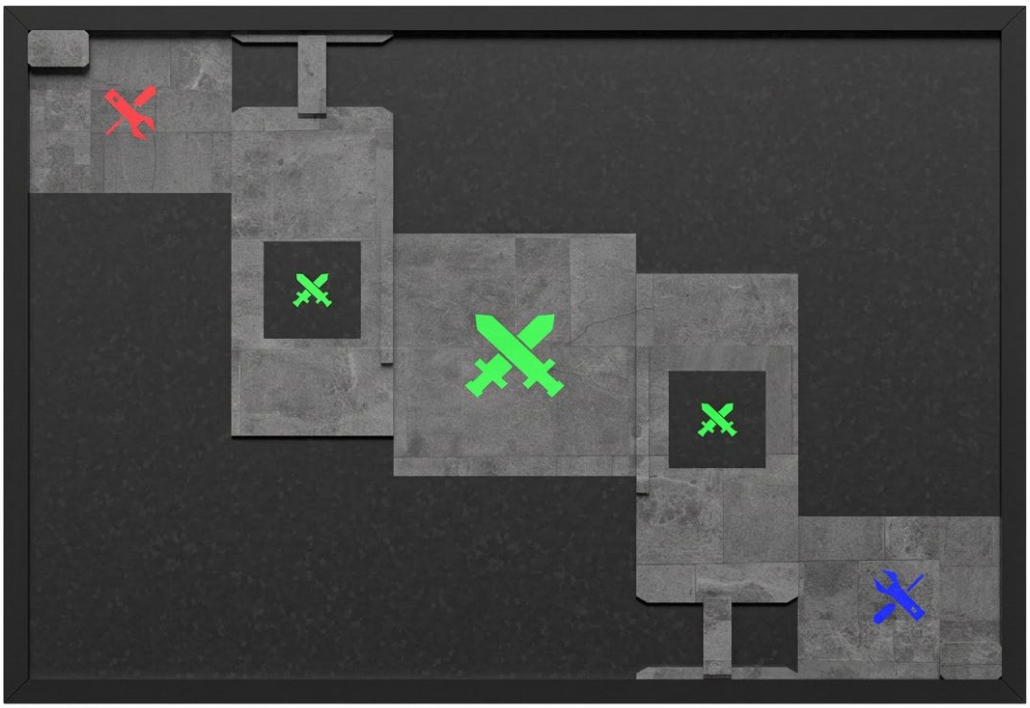


Figure 3-1 Top-view rendering of the 3V3 Match field

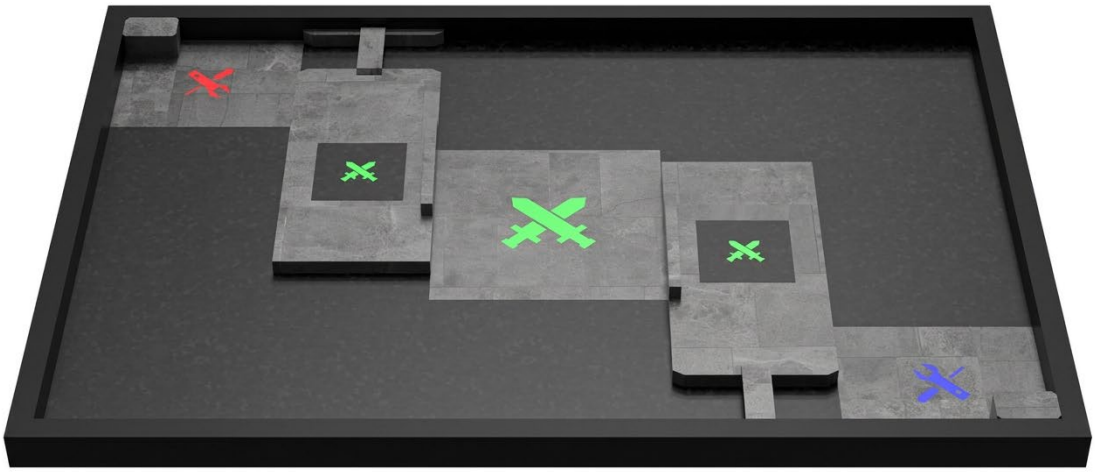


Figure 3-2 Oblique-view rendering of the 3V3 Match field

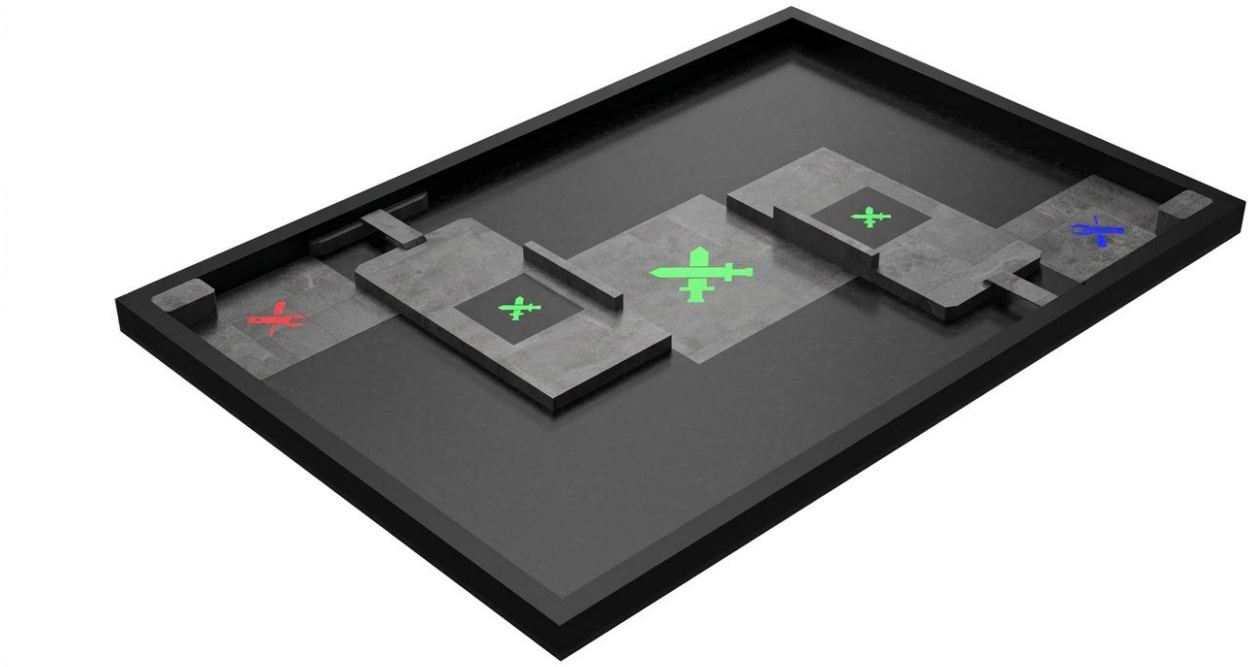
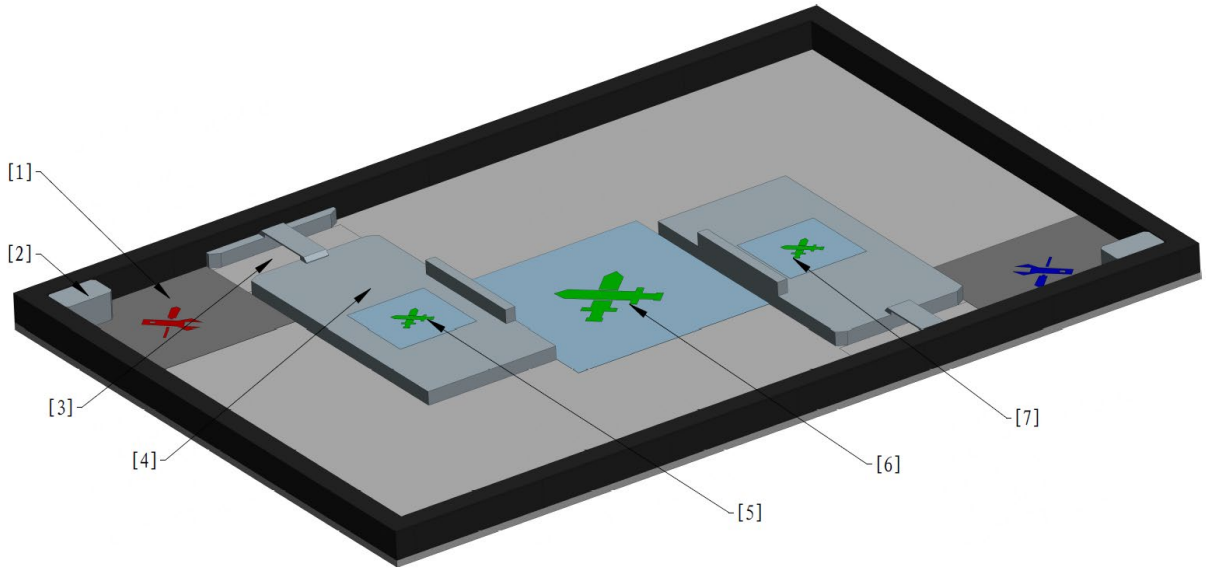


Figure 3-3 Isometric rendering of the 3V3 Match field



[1] Supply area [2] Workbench [3] Tunnel [4] Highland

[5] Red square highland control zone [6] Central control zone [7] Blue square highland control zone

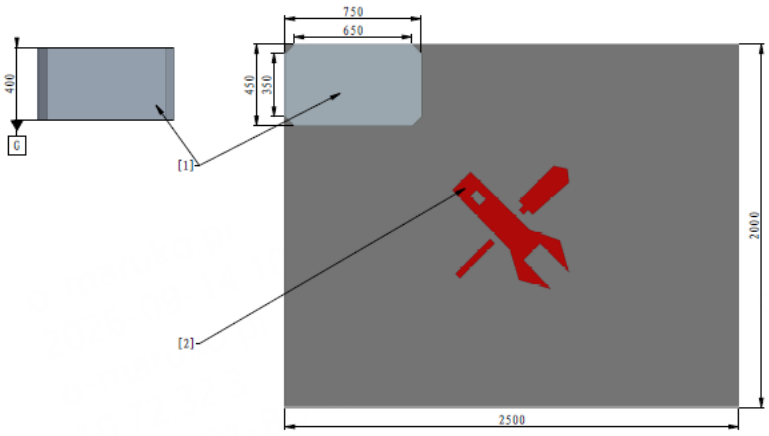
Figure 3-4 Schematic of the 3V3 Match field modules



Figure 3-5 Schematic of the 3V3 Match field dimensions

3.2.2 Starting Zone and Supply Zone

The Starting Zone is where robots are placed before the match officially begins. It also serves as a Supply Zone and contains a workbench. Robots use it to exchange Projectile Allowance, recover HP and install the modular 42mm launching mechanism. Field Interaction Module cards are laid there.



[1] Workbench [2] Supply area

Figure 3-6 Schematic of the Starting Zone and Supply Zone

One side's Supply Zone is a Restricted Zone for the opposing side's robots.

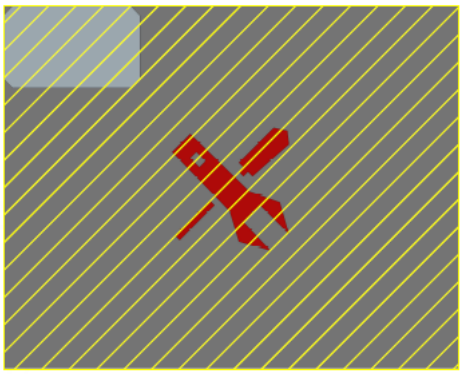
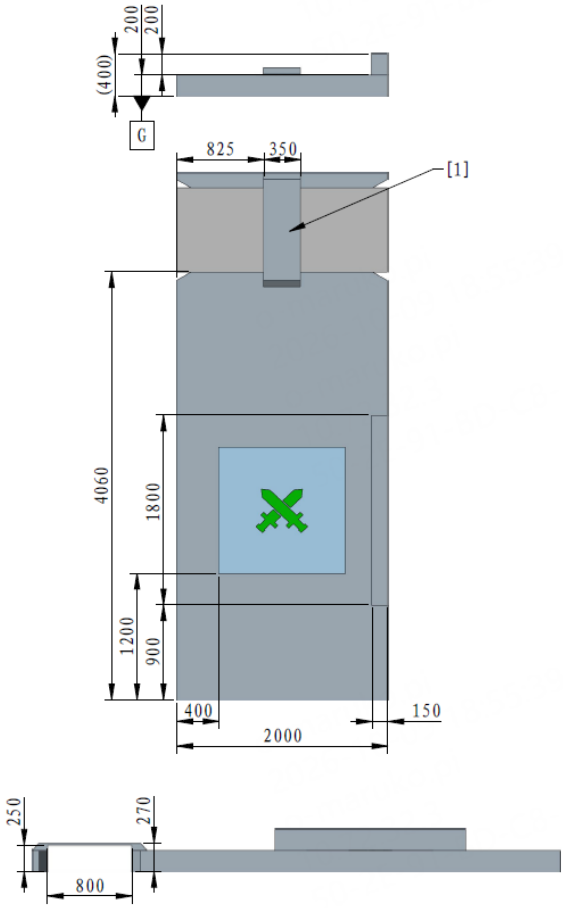


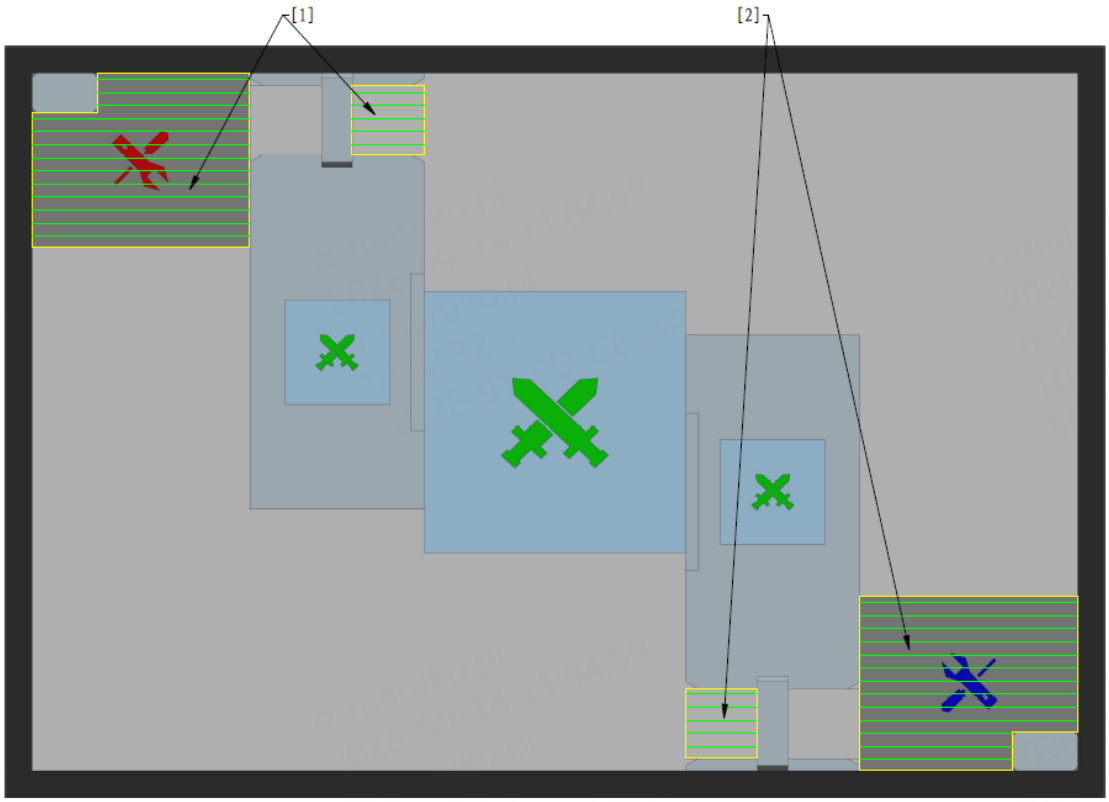
Figure 3-7 Schematic of the Starting Zone and Supply Zone Restricted Zone

3.2.3 Highland and Tunnel



[1] Metal plate

Figure 3-8 Schematic of the Highland

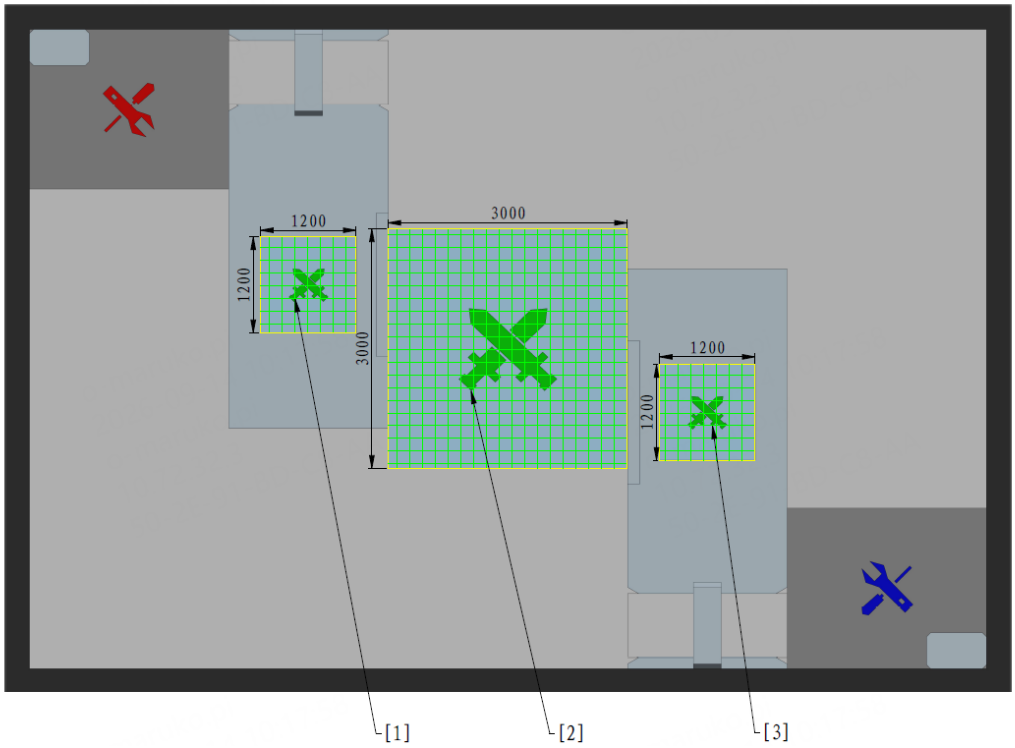


[1] Red tunnel Buff Point [2] Blue tunnel Buff Point

Figure 3-9 Schematic of the Tunnel

3.2.4 Control Zone

The control zone is located in the center of the Highlands and Battlefield, where Field Interaction Module cards are laid.

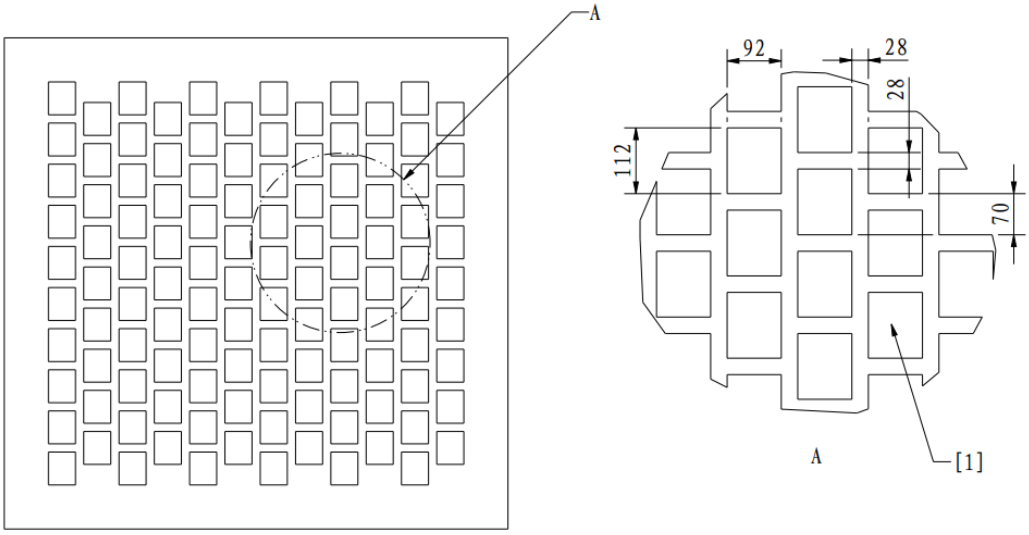


[1] Red square highland control zone

[2] Central control zone

[3] Blue square highland control zone

Figure 3-10 Schematic of the Control Zones



[1] Field Interaction Module card mounting position

Figure 3-11 Field Interaction Module card laying grid

- Field Interaction Module cards are laid at all Battlefield Buff Points, and the cards may have dead zones.



- On top of the Field Interaction Module cards in the Control Zones, Supply Zones and Tunnels, an additional layer of floor covering is laid over the floor covering already present on the field.

3.2.5 Other

3.2.5.1 Projectiles

Robots can attack Armor Modules by launching projectiles and thereby cause damage. Projectile parameters and arrangements for the match are as follows:

Table 3-6 Projectile parameters and usage arrangements

Type	Appearance	Color	Size	Mass	Shore Hardness	Material	Usage
42mm projectile	Similar in shape to a golf ball	White	42.5mm±0.5mm	41g±1g	90±5A	Plastic, TPE	3V3 Match
17mm projectile	Spherical	Yellow-green	16.8mm±0.2mm	3.2g±0.1g	90±5A	Plastic, TPU	3V3 Match, Infantry Match

3.2.5.2 Operator Room

The Operator Room is equipped with a corresponding number of computers, and each computer is equipped with a corresponding display screen, mouse, keyboard, camera image transmission module (receiving end) and its bracket, a 1GbE (RJ45) network port and other official equipment. The camera image transmission module (receiving end) bracket is fixed on the desktop, and the operator can operate the camera image transmission module (receiving end) within the movement range of the bracket.

- Removing the camera video transmission module (receiver) from its bracket is prohibited; the participating team bears the consequences.



- A robot entering the Battlefield automatically connects to the camera video transmission module (receiver) in the Operator Room. A receiver supplied by the participating team cannot be used. The Operator Room receiver's joystick defaults to centered, and its switches default to position C.

3.3 Match Mechanisms

3.3.1 HP Deduction and Exceeding-the-Limit Penalty Mechanisms

3.3.1.1 Attack Damage or Collision

The Armor Modules of the Heavy Robot, Infantry Robot and Sentry Robot may lose HP when attacked or struck. The specific mechanisms are as follows:

Armor Module can detect projectile attacks with a minimum detection interval of 50ms (when using a 42mm projectile to hit the Armor Module, the detection interval may be extended to up to 200ms).

A projectile must strike the Armor Module’s hit surface at a certain speed to be detected. The effective detection speed ranges for different projectile types are shown in the following table:

Table 3-7 Effective detection speeds of Armor Modules for different projectile types

Armor Module	17mm projectile	42mm projectile
Small Armor Module	Greater than 12m/s	Greater than 8m/s



- In the actual competition, because the projectile speed attenuation and the incident angle are not normal to the Armor Module's attack surface, the normal speed of the projectile that touches the Armor Module's attack surface is different from the projectileInitial Launch Speed. Damage detection is based on the normal component of the velocity of the projectile contacting the attacked surface of the Armor Module.
- When one Heavy Robot does not fire 42mm projectile for 4 consecutive seconds, all armor of the opponent's robot will block 42mm projectile damage until the Heavy Robot fires 42mm projectile.
- The Referee System rounds HP to the nearest integer when settling it.

When the robot's Armor Module is hit, there is a probability that it will be judged as projectile damage, but it is not allowed to cause damage to the opponent through impact (including collision with the robot, throwing objects, etc.).

Without any buffs, the raw damage values are as shown in the table below:

Table 3-8 Attack damage HP deduction mechanism

Damage type / Target	42mm projectile	17mm projectile	Ramming
Robot Armor Module	200	20	2

3.3.1.2 Initial Launch Speed Exceeding the Limit

When the Initial Launch Speed of a Heavy Robot, Infantry Robot or Sentry Robot exceeds the limit, that robot's launching mechanism is locked and unlocked after a certain period of time. The time periods are not cumulative.

The locking and unlocking caused by Initial Launch Speed exceeding the limit are independent of the locking and unlocking caused by other conditions. Set the upper limit of the robot's Initial Launch Speed to V0 (m/s), and the Referee System detects that the projectile's actual Initial Launch Speed is V1 (m/s). The specific locking time is as shown in the table below:

Table 3-9 Penalty mechanism for Initial Launch Speed exceeding the limit

17mm projectile	42mm projectile	Lock time
$0 < V_1 - V_0 < 5$	$V_0 < V_1 \leqslant 1.1 * V_0$	15 s
$5 \leqslant V_1 - V_0 < 10$	$1.1 * V_0 < V_1 \leqslant 1.2 * V_0$	20 s
$10 \leqslant V_1 - V_0$	$1.2 * V_0 < V_1$	The rest of the round, with no further unlocking

3.3.1.3 Barrel Heat Exceeding the Limit and Cooling

When the Barrel Heat of a Heavy Robot, Infantry Robot or Sentry Robot exceeds the limit, that robot's launching mechanism is locked. Locking and unlocking caused by excessive Barrel Heat is independent of locking and unlocking caused by other situations. The specific mechanism is as follows:

Let Q0 be the robot's Barrel Heat Limit and Q1 its current Barrel Heat. Whenever the Referee System detects a 17mm projectile, Q1 increases by 10, regardless of the projectile's Initial Launch Speed. Whenever it detects a 42mm projectile, Q1 increases by 100, regardless of that projectile's Initial Launch Speed. Barrel Heat cooling is settled at 10 Hz; the cooling amount per detection cycle equals the per-second cooling rate divided by 10.

When $Q2 > Q1 > Q0$, the image transmission screen of the first-view operation interface of the robot corresponding to the Operator is disconnected, and the launching mechanism will be locked. Until $Q1 = 0$, the first-view operation interface will return to normal and the launching mechanism will be unlocked.

When $Q1 > Q2$, the robot's launching mechanism remains locked for the rest of the current round and is not unlocked again.

For 17mm launching mechanism, $Q2 = Q0 + 100$;

For the 42mm launching mechanism, $Q2 = Q0 + 200$.

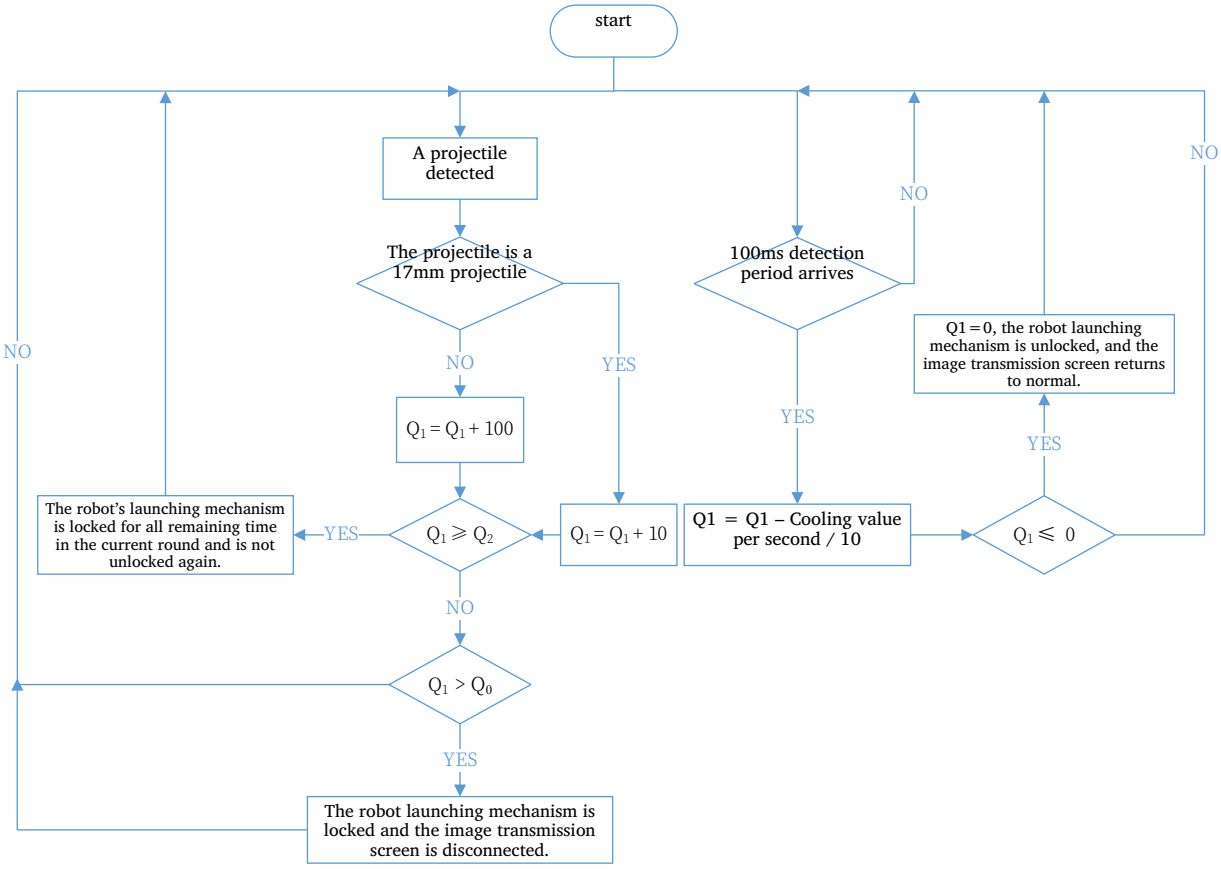


Figure 3-12 Barrel Heat exceeding the limit and cooling mechanism

3.3.1.4 Chassis Power Exceeding the Limit

The Referee System continuously monitors robot Chassis Power, and the chassis must operate within the power limit. When the Chassis Power of a Heavy Robot, Infantry Robot or Sentry Robot exceeds the limit, that robot loses Buffer Energy. If Buffer Energy is depleted while Chassis Power still exceeds the limit, the chassis is powered off for 5 seconds. The Referee System settles Chassis Power detection at a frequency of 10 Hz.

Set Z as the buffer energy, Q as the upper limit of the buffer energy, and it is 60J, Pr as the instantaneous chassis output power, and P1 as the upper limit power.

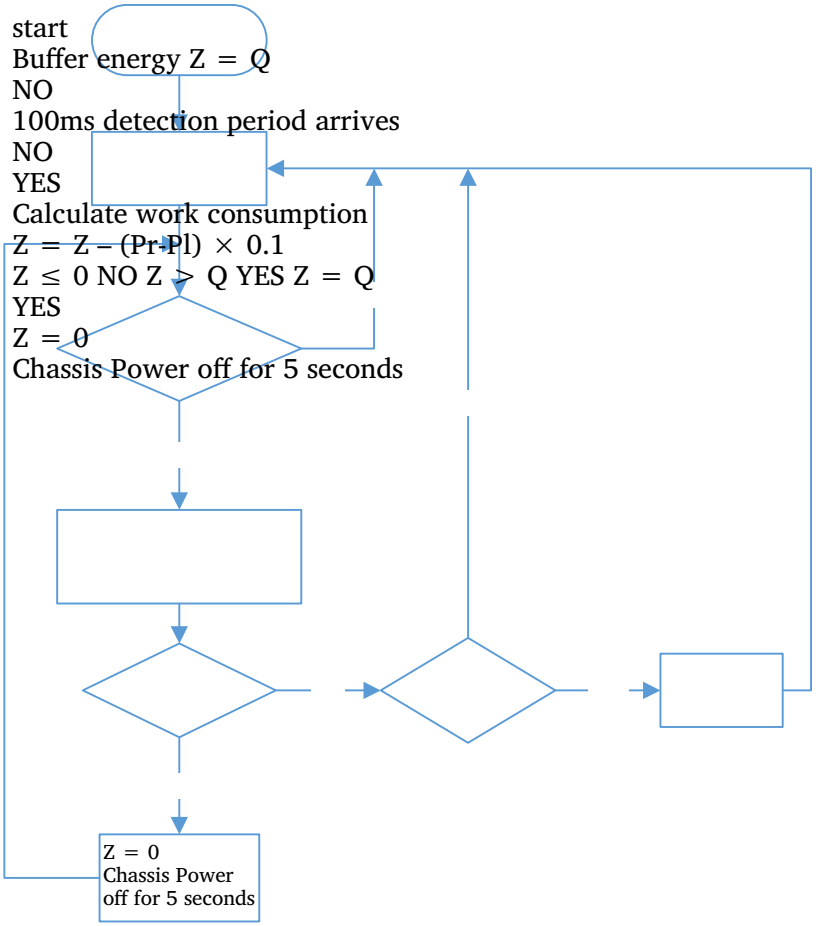


Figure 3-13 Robot chassis detection and Chassis Power exceeding-the-limit penalty mechanism

3.3.1.5 Referee System Onboard Module Offline

Participating teams shall install the Referee System modules corresponding to their robots in accordance with the RoboMaster 2027 University Series Robot Building Specifications Manual;

during the match, they must maintain stable connections between all Referee System modules and the server. The Referee System server checks the connection status of each module at 2 Hz.

3.3.1.5.1 Abnormal Offline

During the competition, if the robot Main Controller Module goes offline and enters the "abnormal offline" state, it can reconnect to the competition.

Table 3-10 Penalty mechanism for robot abnormal offline

Robot type	What to do while offline
Heavy Robot, Infantry Robot and Sentry Robot	□ When the launching mechanism, gimbal, and chassis are powered off, 5% of the upper limit HP will be deducted every second until it reaches 0 □ Field Interaction Module is invalid □ The robot no longer detects damage caused by impact and projectile hits and the module offline HP deduction □ The resurrection reading bar will no longer increase □ Multi-machine communication is disconnected

3.3.1.5.2 Armor Module or Supercapacitor Management Module Offline

If a Heavy Robot, Infantry Robot or Sentry Robot's Armor Module or Supercapacitor Management Module goes offline because of a problem with its own design or structure, the robot loses HP.

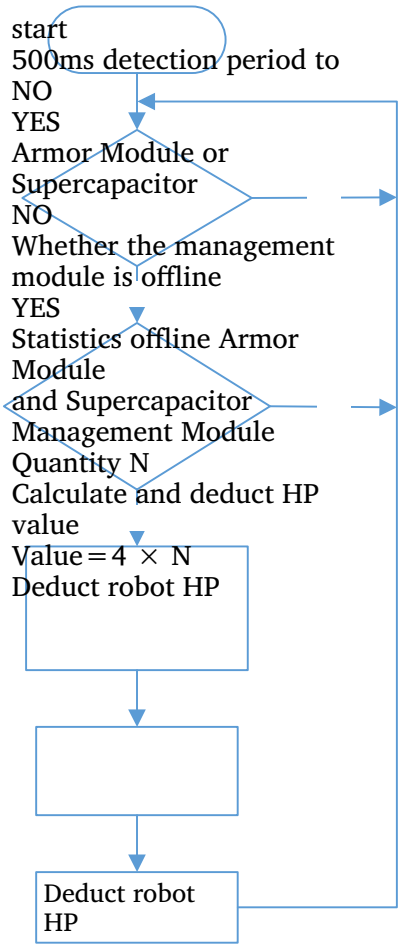


Figure 3-14 HP deduction mechanism for Armor Module and Supercapacitor Management Module offline

3.3.1.5.3 Speed Monitor Module Offline

If the 17mm Speed Monitor Module or 42mm Speed Monitor Module installed on a Heavy Robot, Infantry Robot or Sentry Robot goes offline, the robot's corresponding 17mm or 42mm launching mechanism is powered off immediately.

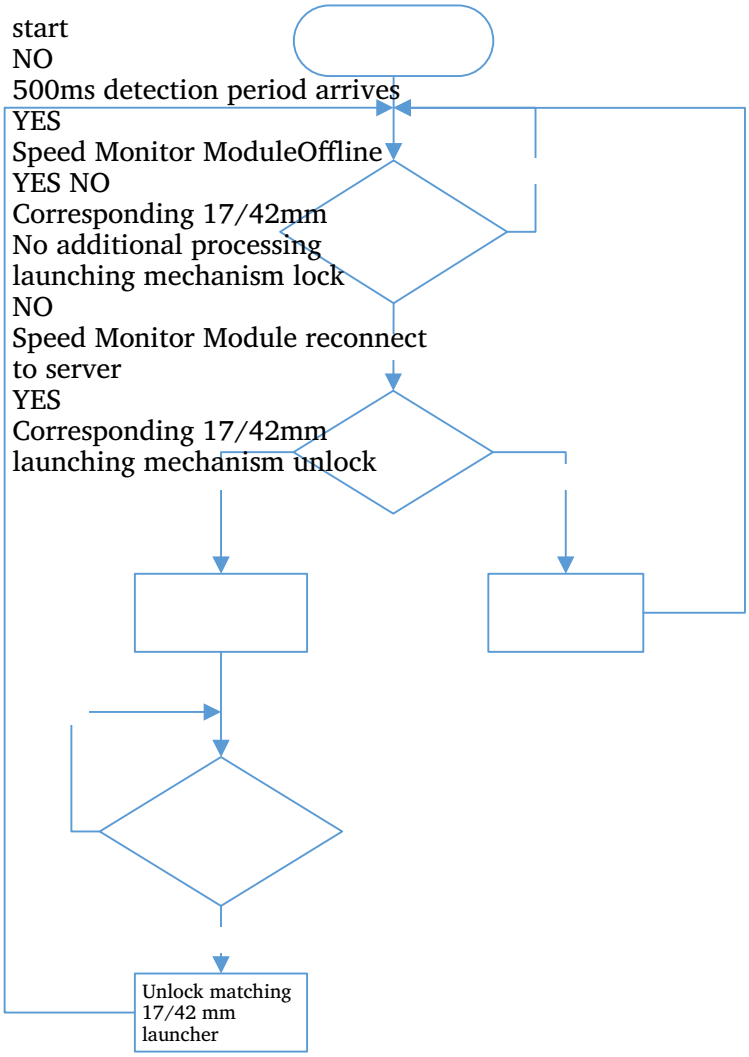


Figure 3-15 Power-off mechanism for Speed Monitor Module offline

3.3.1.6 Penalty Damage

When a Heavy Robot, Infantry Robot or Sentry Robot receives a Yellow Card Warning or Red Card Warning, it may lose HP. See '6.1 Penalty System' for details.

3.3.2 HP Recovery and Revival Mechanism

Except when abnormally offline or Ejected, Ground Robots can recover HP and undergo Revival.

3.3.2.1 HP Recovery Mechanism

A Heavy Robot, Infantry Robot or Sentry Robot can activate its own Supply Zone. Doing so restores HP at a rate of 25% of Max HP per second.

3.3.2.2 Revival Mechanism

When a robot is Defeated, its Barrel Heat resets to 0 and its Buffer Energy resets to the Buffer Energy limit. A Defeated robot can be revived automatically after completing its revival progress. During automatic revival, progress increases by 1 point per second. The revival progress required for a robot's first Defeat is 5; after each subsequent Defeat of the same robot, the required progress increases by 5.

After the robot is resurrected, its HP will be restored to 20% of the upper limit HP, it will be temporarily invincible for 30 seconds, and it will also enter a "weak" state. The robot in the "weak" state has the following characteristics:

- The launching mechanism is locked.
- Unable to activate control zone.

Afterward, when the robot detects the Field Interaction Module card in its own Supply Zone, its Weakened and Invincible states are removed.

3.3.3 Economy System

At the beginning of the game, both parties will receive a certain amount of initial Gold. As the game progresses, both parties will receive Gold regularly. Gold can only be used to exchange for Projectile Allowance. The specific Gold increment rules are shown in the following table:

Table 3-11 Periodic Gold increments for robots

Match countdown	Red Side	Blue Side
04:59	200 (initial)	200 (initial)
03:59	200	200
02:59	200	200
01:59	300	300
00:59	300	300

When the difference in Control Points between the two parties is 200 or 400 for the first time, the party with fewer Control Points will receive 200 Gold. In each game, one side can obtain up to 400 Gold through this mechanism. Regarding the Control Points mechanism, see "3.3.4 control zone and Control Points mechanism" for details.

Table 3-12 Exchange rules

Exchange item	Exchange rate
17mm projectile	10 Gold/10 projectiles
42mm projectile	10 Gold/1 projectile

For every projectile fired by the robot, the Projectile Allowance corresponding to the type of projectile launched is reduced by 1 round.

In the following special situations, all Armor Modules of the opposing robots block 42mm projectile damage until the Heavy Robot is Alive and its Projectile Allowance is greater than 0:

1. 3 seconds after the Heavy Robot changes from Alive to Defeated or becomes abnormally offline.
2. Heavy Robot launches 42mm projectile when Projectile Allowance is 0 (overshooting behavior occurs).
3. After the Heavy Robot has been Defeated, once it launches its third 42mm projectile.
4. The communication packet loss rate between the modular 42mm launching mechanism and the Referee SystemMain Controller Module is greater than or equal to 5%.

17mm projectile damage will not be blocked.

When a Heavy Robot or Infantry Robot activates a Supply Zone, its Operator can redeem 42mm or 17mm Projectile Allowance, respectively, through the Referee System Player Client.

3.3.4 Control Zones and Control Points Mechanism

There are three groups of control zones in the Battlefield, the red highland control zone, the blue highland control zone and the central control zone. After the game starts, the control zone mechanism takes effect.

3.3.4.1 Control Zone Mechanism

At the start of the game, all three control zones are neutral.

Each party has its own independent "occupation progress" count for each control zone. The highest count is 5 and the lowest is 0. When a party's "occupation progress" of a control zone reaches 5, that party "controls" the control zone; when a party's "occupation progress" of a control zone is reduced to 0, that party loses control there. When neither party controls a control zone, the control zone is in a neutral state.

When a Control Zone is activated only by one side’s robot and the opposing side’s occupation progress is 0, the activating side’s occupation progress for that zone increases by 1 per second. When a Control Zone is activated only by one side’s robot and the opposing side’s occupation progress is not 0, the opposing side’s occupation progress for that zone decreases by 1 per second.

When both sides activate the same control zone, the change in occupation progress is suspended until only one robot remains active in the control zone.

When neither party activates a non-neutral control zone, both parties' occupation progress of the control zone will be reduced by 1 per second.

3.3.4.2 Control Points Mechanism

When the game starts, both sides have 600 Control Points:

- For each Control Zone one side controls, the opposing side loses 1 Control Point per second.
- Each time a robot on one side is Defeated, that side loses 10 Control Points.

When the difference in Control Points between the two sides is at least 200, the side with fewer Control Points needs 33% less Revival progress and its HP Recovery rate in the Supply Zone increases to 33%/s.

When the difference in Control Points between the two sides is at least 400, the side with fewer Control Points needs 50% less Revival progress and its HP Recovery rate in the Supply Zone increases to 50%/s.

3.3.5 Field Buffs Mechanism

3.3.5.1 Highland Buff Point and Highland Control Zone Mechanism

Robots on both sides can activate both sides' Highland Buff Points, and multiple robots on the same side can activate a Highland Buff Point simultaneously.

Robots that activate Highland Buff Point gain a 25% defense boost.

The highland Buff Point is also a highland control zone. For details on the control zone mechanism, see "3.3.4 Control Zone and Control Points Mechanism".

3.3.5.2 Tunnel Buff Point Mechanism

Each robot can activate its own tunnel Buff Point. Multiple robots on the same side can activate tunnel Buff Points at the same time.

When the same robot detects Field Interaction Module cards at two tunnel Buff Points in a Battlefield within 3 seconds, and no other Field Interaction Module cards are detected on the way, it can obtain a 1.5x heat cooling gain for 60 seconds.

3.3.5.3 Control Zone Buff Point Mechanism

Robots on both sides can activate Control Zones, and multiple robots on the same side can activate a Control Zone simultaneously. For the specific effects, see '3.3.4 Control Zones and Control Points Mechanism'.



After the robot does not detect the Field Interaction Module card of Buff Point, the activation status will expire after 2 seconds.

3.3.6 Heavy Robot Special Mechanism

At the start of a match, the Heavy Robot's performance attributes are low. To improve them, it must install the modular 42mm launching mechanism and launch a 42mm projectile correctly detected by the Referee System. For details on performance attributes, see '3.1.1 Heavy Robot'.

The Heavy Robot can perform the following operations, but only at the Workbench:

Decompose into sub-robots or multiple sub-systems connected with flex cables, or merge from above.

Install or remove the modular 42mm launching mechanism.

When the Heavy Robot has not successfully completed installation of the modular 42mm launching mechanism and activates its own Supply Zone, it is Invincible.

3.3.7 Competition Format and Victory Conditions

The 3V3 competition is divided into group round-robin and knockout. The format of the group round robin is BO2 and the knockout format is BO3.

The following are the winning conditions for a single game:

1. When one side's Control Points reach 0, the round ends immediately and the side whose Control Points are not 0 wins.
2. When the Round time expires and neither side's Control Points are 0, the side with more Control Points wins.
3. When the Round time expires and both sides have the same nonzero Control Points, the side with greater total Attack Damage wins.
4. When the Round time expires and both sides have the same nonzero Control Points and equal total Attack Damage, the side with greater total remaining HP across its robots wins.
5. If the above conditions cannot determine victory, the game will be deemed a draw. In the event of a tie in the knockout rounds, an extra round will be played immediately until a winner is determined.

4 Infantry Match

During the two-minute Round, the robots of the two teams engage in a shooting confrontation in the core competition area, the Battlefield. The side that destroys the opposing robot wins the match.

4.1 Robots and Operators

The lineup of robots and operators is as follows:

Table 4-1 Infantry Match robot and Operator lineup

Type	Number	Main lineup quantity	Maximum number of reserve robots quantity	Maximum number of equipped Custom Terminals	Maximum number of reserve Custom Terminals	Operator Lineup
Infantry Robot	3	1	0	1	0	1 person per robot

- Operator can only be held by Battlefield personnel other than the supervising teacher in the current participating team.
- After each game, the Operator can be replaced from the current Battlefield personnel except the supervising teacher.

4.1.1 Infantry Robot

The Infantry Robot can launch 17mm projectiles.

Table 4-2 Key characteristics of the Infantry Robot

Key characteristic	Description
Initial zone	Starting Zone
Operating mode	Unlimited, with at most 1 Remote Controller and 1 Custom Terminal configured
Multi-robot communication	Allowed
Launching mechanism	1 17mm launching mechanism
Initial Launch Speed limit (m/s)	25
Performance	- Initial HP/Max HP is 200 • Barrel Heat Limit is 80

Key characteristic	Description
	Barrel Cooling Rate is 12/second The upper limit of Chassis Power is 120W
HP deduction and limit-exceeding penalty mechanism	- Initial Launch Speed exceeding the limit: applies • Barrel Heat exceeding the limit and cooling: applies • Chassis Power exceeding the limit: applies • Attack damage or collision: applies • Referee System onboard module offline: applies • Penalty Damage: applies See "3.3.1 HP Deduction and Limit-Exceeding Penalty Mechanism".
HP Recovery and Revival mechanism	Not applicable
Projectile loading and resupply	- During the two-minute Setup Period, 17mm projectiles can be preloaded. • During the two-minute Round, the initial Projectile Allowance is 200 projectiles and cannot be increased during the round.

4.1.2 Custom Terminal

Custom Terminal (original custom controller and custom client) is a set of multi-purpose equipment self-made by the participating team. It is used to monitor the motion and status of the robot, control the robot's actions, receive game-related information and robot image transmission from the Referee System event engine server, and can send event-specific interactive instructions to the event engine server to expand the possibility of human-computer interaction.

A typical Custom Terminal can be composed of a display device, a computing platform (including memory, controller and arithmetic unit) and a set of input devices; it can also be integrated, such as a handheld tablet device with a network port.

Example: A team created a Custom Terminal, which can receive the robot's image transmission and display it on a self-made display. At the same time, a self-made UI displayed through an unofficial link is superimposed, and the robot can be controlled through a matching controller. As a result, the team can complete the game without using the official Referee System player terminal and Remote Controller.

All Custom Terminals in the Operator Room can be wired to connect and communicate with each other, and the Custom Terminals can be wired to the event engine server and send and receive relevant content through the RJ45 interface set in the Operator Room.

Custom Terminal can receive event information displayed on the Referee System player terminal and accessible to oneself, such as: HP of both parties, one's own economy, remaining time of the game, and key events of the game, etc.

A Custom Terminal can send match instructions supported by the Referee System player terminal, such as selecting a Performance System before a match or redeeming Projectile Allowance during a match,

and other active commands.

4.2 Competition Area

The core competition area of the Infantry Match is called the ‘Battlefield’. It is a 5 m long by 5 m wide area containing separate Starting Zones and bunkers for the Red and Blue sides. For details on projectiles and the Operator Room, see ‘3.2.5.1 Projectiles’ and ‘3.2.5.2 Operator Room’.

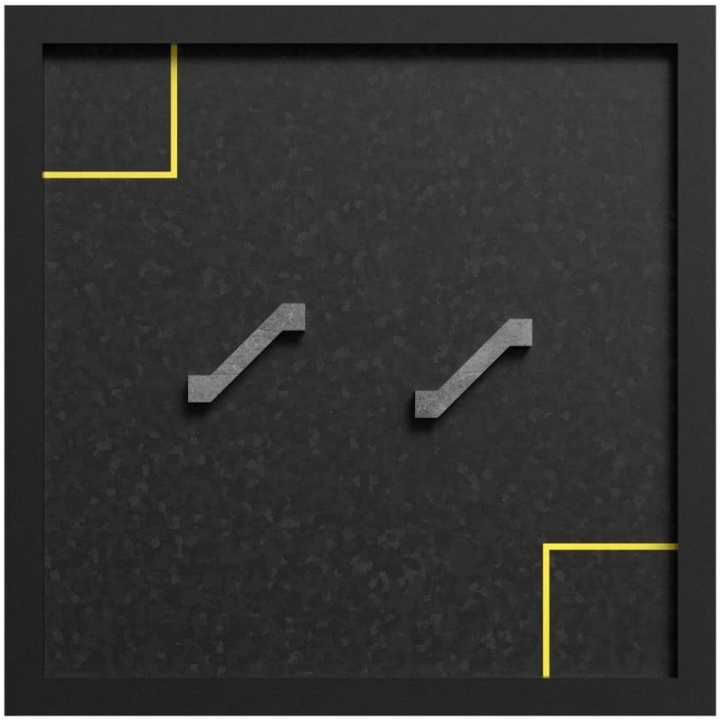


Figure 4-1 Top-view render of the Infantry Match field

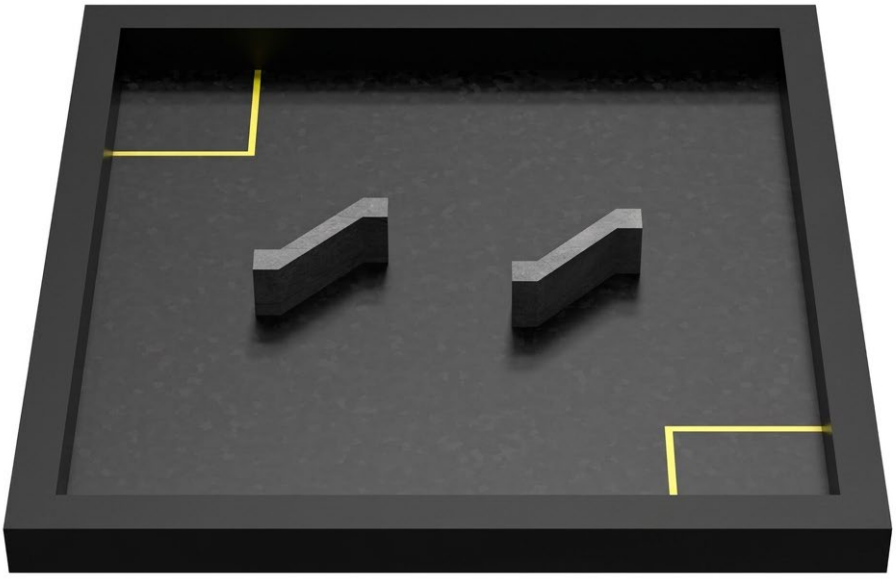


Figure 4-2 Oblique-view render of the Infantry Match field

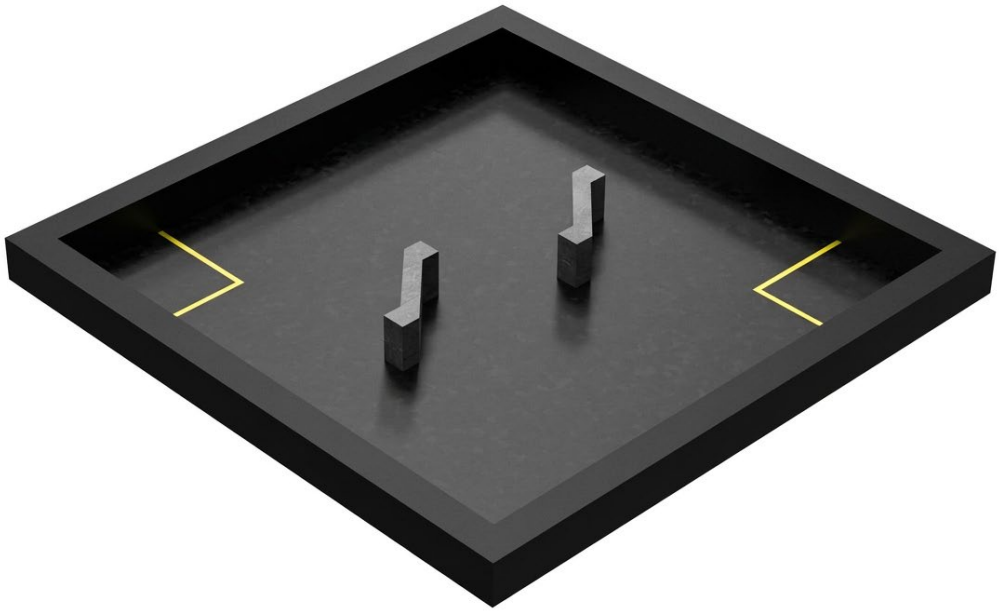
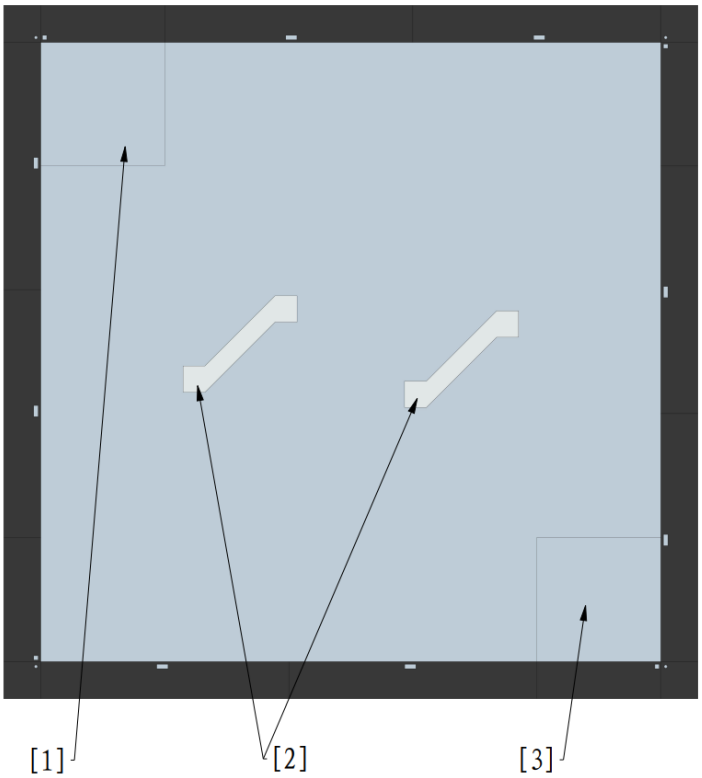


Figure 4-3 Isometric render of the Infantry Match field



[1] Red team starting area [2] Bunker [3] Blue team starting area

Figure 4-4 Top view of the Infantry Match field

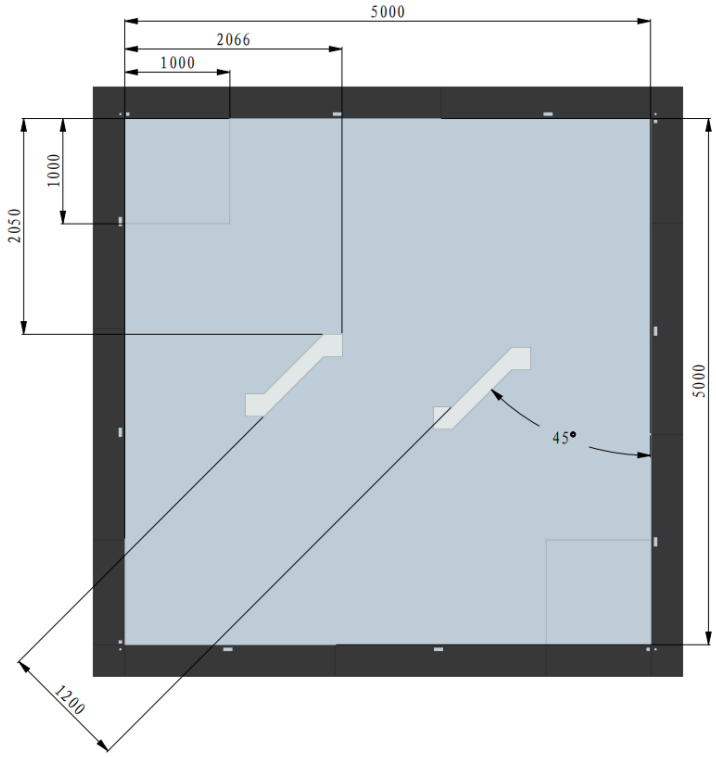


Figure 4-5 Dimension diagram of the Infantry Match field

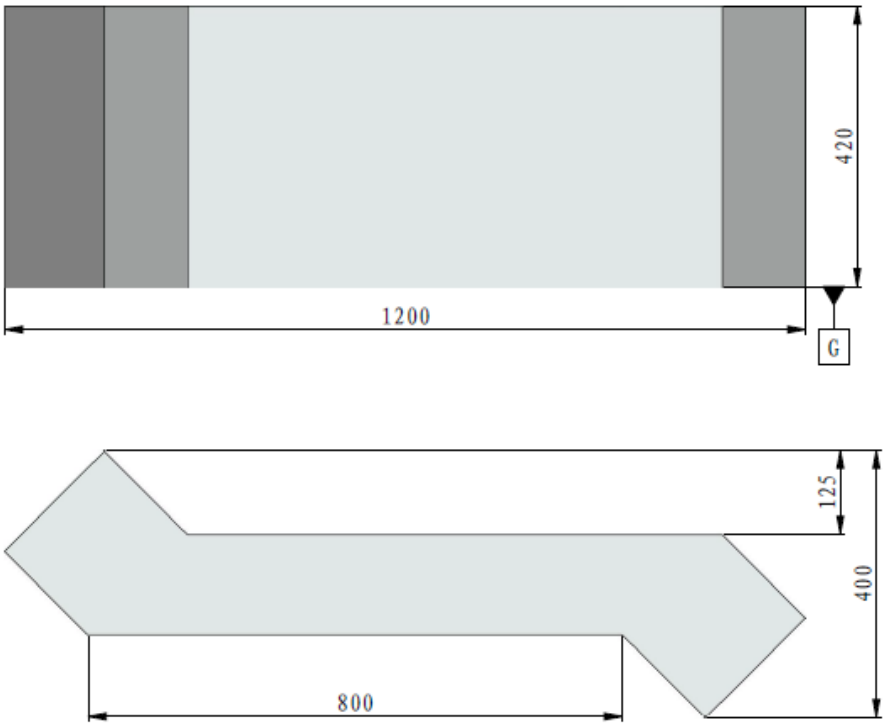


Figure 4-6 Cover diagram

4.3 Match Mechanisms

For the Infantry Robot details, see ‘3.3.1 HP Deduction and Over-Limit Penalty Mechanism’.

4.3.1 HP Recovery Mechanism

In the Infantry Match, each time a robot deals projectile damage to the opponent, it recovers HP equal to 50% of the damage dealt.

4.4 Competition Format and Victory Conditions

The Infantry Match has two stages: a group round-robin stage and a knockout stage. The group round-robin format is BO2, and the knockout format is BO3.

The following are the winning conditions for a single game:

1. When the HP of one robot drops to 0, the game ends immediately and the opponent wins.
2. When the game time runs out, and neither side's HP is 0, the robot with the higher remaining HP wins.
3. When the time of a game runs out, the HP of both sides is not 0, and the remaining HP of the robots is the same, the side whose HP was deducted later for the first time wins.
4. When the time of a game runs out, and both sides' HP is not 0, and both sides are at full health, the side with the higher total amount of ammunition will win.
5. When the time of a game runs out, if the HP of both sides is not 0, both are full health and the total amount of bullets fired is the same, the side that fired the first bullet earlier wins.
6. If the above conditions cannot determine victory, it will be considered a draw.

4.4.1 Special Cases in the Infantry Match

1. If the score between two teams in the knockout stage is 0:1 or 1:0, the one who wins 1 game will win.
2. If two teams in the knockout stage have a score of 0:0 or 1:1, compare their time-averaged net remaining HP across the rounds completed to date; the team with the higher value wins.



- If the divisional Infantry Match schedule includes two group-stage rounds, compare only the teams’ average net remaining HP per round from the start of the second group-stage round through the matches completed to date.
- If two teams have the same time-averaged net remaining HP and the winner cannot be determined, the Organizing Committee will determine the winner by an on-site draw to ensure that the subsequent schedule proceeds smoothly.

5 Match Procedure

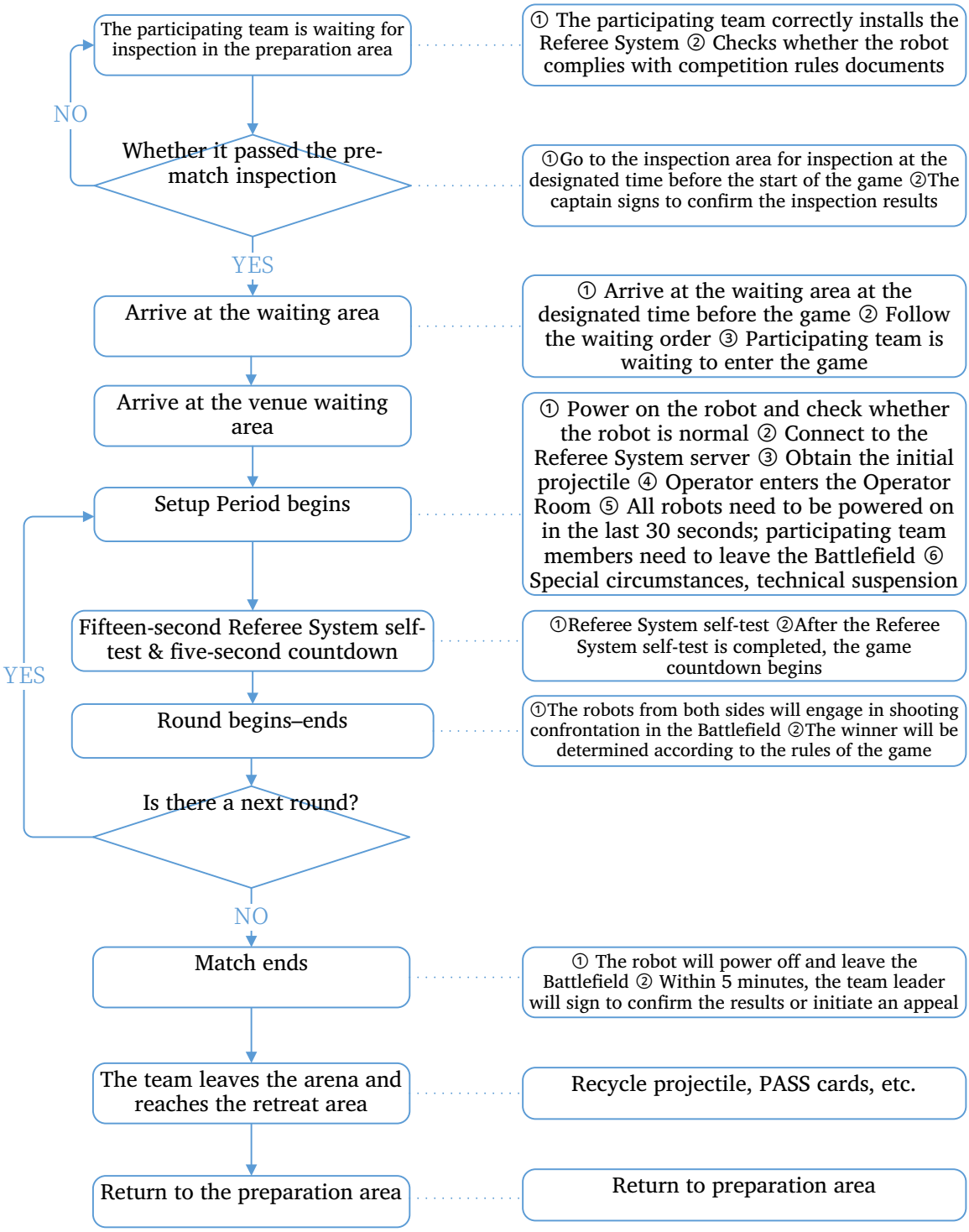


Figure 5-1 Single-match flowchart

5.1 Pre-Match Inspection

- The inspection results of Pre-Match Inspection and adaptive training are for reference only and do not serve as the basis for passing the inspection for the official competition.
- The inspection results of official matches are only effective for that match.
- Passing inspection only proves that the robot met the inspection pass criteria at the time of inspection; the team must itself ensure that the robot always meets the Robot Building Specifications Manual requirements. During inspection, all robot functions must be proactively demonstrated to the inspectors.

To ensure that the robot produced by the participating team meets the requirements of the "RoboMaster 2027 University Series Robot Building Specifications Manual"

For Pre-Match Inspection, teams in the 3V3 Match and Infantry Match must arrive at the inspection area 60 minutes and 40 minutes in advance, respectively. The inspection process is as follows:

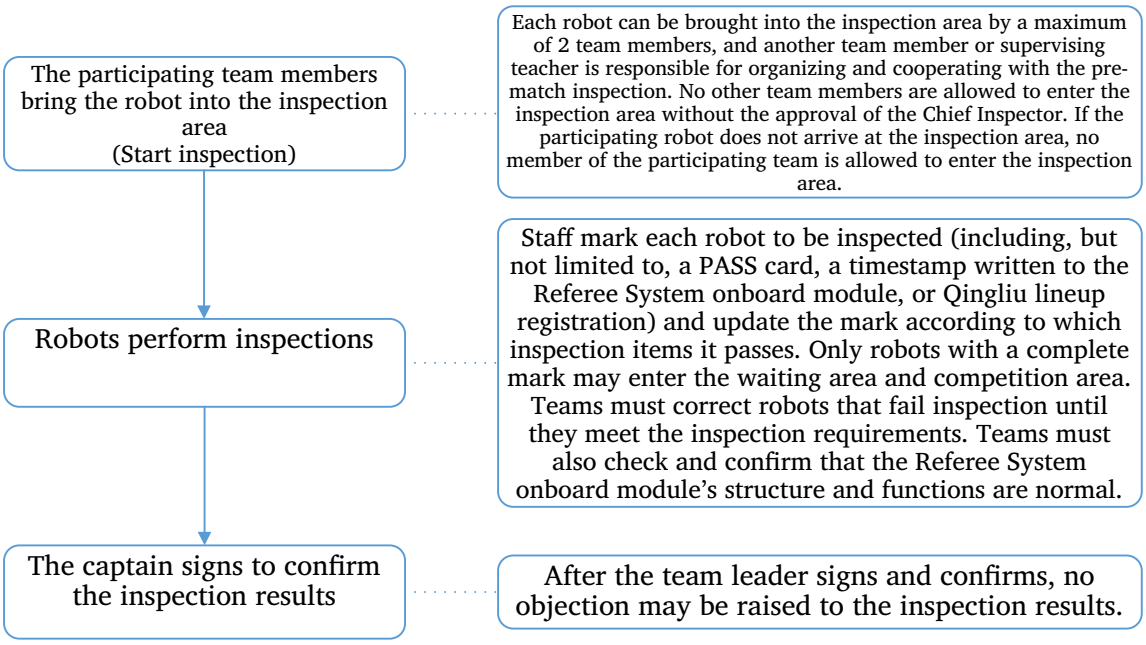


Figure 5-2 Pre-Match Inspection flowchart

In RMUL's 3V3 competition, each participating team can borrow the Referee System of at most one spare robot.

5.2 Waiting Area

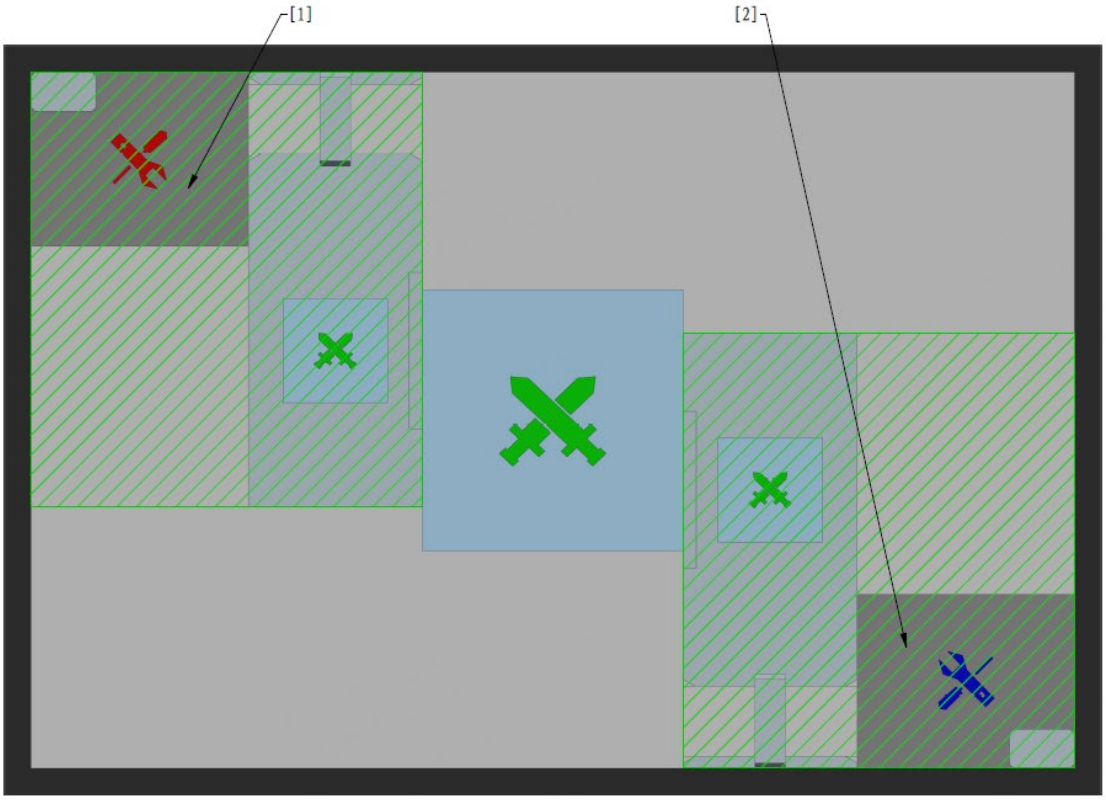
After completing Pre-Match Inspection, a participating team must arrive at the waiting area at least 10 minutes before each match. In the 3V3 Match, each team may have at most 10 field personnel; in the Infantry Match, at most 4. One field member must wear the 'Captain' armband and perform the captain's duties. If the team needs to repair a robot after entering the waiting area, it must obtain referee approval. The robot may leave the waiting area for repair only after waiting-area staff remove its PASS card. After repair, the robot must return to the inspection area and pass Pre-Match Inspection again before returning to the waiting area. If the robot is late to the waiting area as a result, it may not compete, and the participating team bears the consequences.

‘Captain’ armband: Any field team member wearing a ‘Captain’ armband performs the captain’s duties during the match. The captain manages the team’s match procedure, confirms results, submits Team Technical Timeout requests, files Appeals, and so on.

After the team is sent from the waiting area to the field, it enters the field’s waiting zone and places its robots there. Once the referee approves, the team carries the robots to the Battlefield entrance and stands by. The referee opens the door and guides team members onto the field according to the match procedure. The Setup Period countdown starts as the door opens.

5.3 Setup Period

The Setup Period is 3 minutes for the 3V3 Match and 2 minutes for the Infantry Match. During it, field team members place robots in their respective Starting Zones, check that the Referee System is operating normally, preload projectiles and debug the modular 42mm launching mechanism. They may repair robots or replace equivalent parts in accordance with the specifications. In the 3V3 Match, they must debug robots in their side’s designated debugging zone; in the Infantry Match, they may debug only inside the Starting Zone. Each team may have only one field team member leave the designated debugging zone to debug.



[1] Red team designated debugging area [2] Blue team designated debugging area

Figure 5-3 Diagram of the two sides' designated debugging zones

- Equivalent parts: standard modules or parts with the same material, shape and function, such as model motors and self-made friction wheel modules

and so on. The main structure that carries the robot body, such as its chassis, cannot be replaced by an equivalent part.

- If the ‘equivalent parts’ include Referee System onboard modules, the participating team must ensure that those modules function normally (including version, sensor calibration and installation specifications). An abnormality in an onboard module included in the ‘equivalent parts’ does not trigger an Official Technical Pause.

When there is 1 minute and 30 seconds left in the preparation phase, it is recommended that the Operator enter the Operator Room to complete the debugging of the keyboard and mouse, and check to confirm that the robot control and official equipment are operating normally. If the official equipment cannot operate normally, Battlefield personnel must request it before there are 15 seconds left in the preparation phase, otherwise the referee will not grant a technical timeout. Except for the Operator corresponding to the robot on the field, other Battlefield personnel are not allowed to enter the Operator Room.

When there are 30 seconds left in the preparation phase, all robots in the Battlefield need to be powered on, unpowered robots need to be moved out of the Battlefield, and everyone in the Battlefield must leave in an orderly manner.

At the end of the Setup Period, field personnel must place the Sentry Robot’s debugging Remote Controller in the designated area at the Battlefield entrance.

5.3.1 Official Technical Timeout

During the preparation phase, when the Referee System or Operator Room equipment fails (fault conditions are shown in the table below) or when the robot needs to be temporarily inspected, the Head Referee can initiate an official technical pause and pause countdown. The suspension initiation time is determined by the Head Referee depending on the situation.

During an Official Technical Pause, field team members must cooperate with the referees to resolve faults related to the Referee System or official equipment, and may not repair other faults.

If the Referee System fails and must be replaced, the replacement must comply with the time limit for replacing the Referee System onboard module in the RoboMaster 2027 University Series Robot Building Specifications Manual.

After the Referee System or official equipment fault is resolved and the Head Referee resumes the countdown, field team members must follow the three-minute Setup Period requirements and leave the Battlefield at the specified time.

Table 5-1 Failure situations

Item	Description
1	An official device in the Operator Room fails, or key match props in the Battlefield suffer structural damage or functional abnormality.
2	During the first round's Setup Period, a Referee System onboard module fails, for example: Armor Module damaged, Speed Monitor Module offline, etc.
3	During the Setup Period, the Referee System Main Controller cannot connect to the server, or the robot cannot properly transmit video back to the Operator Room.
4	Other situations that the Head Referee determines require an official technical timeout.

Because it is impossible to determine whether a failure is in the Referee System onboard module itself, caused by defects in the participating robot’s electrical or structural design, or caused by robot combat in an earlier match, Routine Battle Damage—except for the Regulation 3 case above—does not trigger an Official Technical Pause. The Referee will provide a spare Referee System onboard module. The participating team may request a Team Technical Pause to repair the robot.

If, after investigation by the referee, the fault conditions described in Regulations 2 and 3 above are caused by the participating team, the referee will explain the situation and end the official technical suspension.

5.3.2 Team Technical Timeout

If the robot's mechanical structure or software system, or the keyboard, mouse or other equipment brought into the Operator Room, fails, the participating team may request a Team Technical Pause only before the Setup Period's 15-second countdown. The captain must apply to a referee in the Battlefield or Operator Room and state the requested pause duration and reason. Once the request is made and communicated to the Head Referee, it cannot be canceled or changed.

Once the Head Referee confirms a Team Technical Pause, the Referee notifies both teams at the same time, regardless of which side requested it.

After the Head Referee confirms that both teams are ready, the technical pause may end early. Even if a team does not enter the Battlefield or the pause ends early, the opportunity consumed is still the one corresponding to the duration declared by the team when it applied.

To ensure that the subsequent schedule runs on time, both sides together may initiate only 1 Team Technical Pause during the same Setup Period, on a first-come, first-served basis. The post-match result confirmation form records whether any technical-pause opportunity was consumed during the match.

In both the 3V3 Match and Infantry Match, each participating team has two opportunities to request a 2-minute Team Technical Pause.

5.4 15-Second Referee System Self-Check Period

After the Setup Period, the match enters the 15-Second Referee System Self-Check Period. During the self-check, the Referee System server automatically checks the Referee System Client connection, the robots' Referee System onboard module status and the Field Props status, and resets all robots' HP to ensure they start the match at full HP.

5.5 Five-Second Referee System Self-Check Period

After the 15-Second Referee System Self-Check Period, the match enters the Five-Second Countdown. During this time, the Referee System Client does not respond to robot control commands, including those from Custom Terminals. When the countdown ends, the Client resumes responding to control commands and the match starts immediately.

5.6 Round Stage

Robots engage in shooting combat in the core competition area, the Battlefield.

5.7 End of Match

A Round ends when its time expires or a team triggers a Victory Condition early. A Match ends when a winner is determined or all its Rounds have been completed.

5.8 Result Confirmation

For each Match, the Referee records on the result confirmation form the main penalties in each Round, key settlement information at the end of the match, the result and the use of Team Technical Pause opportunities.

The captain must sign to confirm the match result within 5 minutes after the Match ends. If the captain neither signs nor files an Appeal within 5 minutes, the result is deemed confirmed.

5.9 Exit

After a 3V3 Match, the participating team must power off its robots, carry them off the Battlefield and take them to the projectile-clearing area. There, following staff instructions, it must return all armbands and PASS cards, remove all projectiles from the robots, and return all projectiles used in the match. After an Infantry Match, the team must carry out these exit procedures on the field.

6 Violations and Penalties

To ensure fair competition and discipline, participating teams, participants and robots must strictly follow the competition rules. Referees issue appropriate penalties for violations. Some penalties issued before the match officially starts take effect after it starts.

Major penalties and all appeals during the game will be announced.

The penalties corresponding to all the violations in this chapter will be determined by the Head Referee based on the actual situation of the game. If there are situations during the game that affect the fairness of the game but are not covered by the penalty details and serious violations, the Head Referee will make a decision based on the actual situation.



If one party's behavior directly causes the other party to commit a violation, it will not be regarded as a violation by the other party, but the other party must terminate the violation in a timely manner.

6.1 Penalty System

6.1.1 Penalty Methods

During the event, the Referee System or referee will penalize participants and robots that do not comply with the rules of the event. The penalties are as follows:

Table 6-1 Penalty Methods

Penalty Method	Description
Automatic Referee System penalty	In "3.3.1 HP Deduction and Exceeding-the-Limit Penalty Mechanism", all penalties other than attack damage or collisions and Penalty Damage are automatic Referee System penalties.
Manual Referee System penalty	Penalties issued by referees operating the Referee System server for robot violations.
Manual referee penalty	Applies to scenarios where the Referee System cannot be used to issue penalties, such as verbal warnings and disqualification from the competition.

6.1.2 Penalty Types

The types of manual penalties during the game are as shown in the following table:

Table 6-2 Penalty Types

Penalty Type	Description
Verbal Warning	Verbal warning
Yellow Card Warning	- One-side Yellow Card:

Penalty type	Description
	- If the violating robot is a Sentry Robot, its chassis power is cut for 2 s, and the operator interfaces of the other robots are obscured for 2 s. - If the violating robot is not a Sentry Robot, its operator interface is obscured for 5 s, and the operator interfaces of the other robots are obscured for 2 s. - The Referee System automatically deducts 15% of the violating robot's current Max HP, and the other surviving robots each have 5% of their current Max HP deducted. If a robot receives another Yellow Card Warning within 30 s after a Yellow Card Warning, the percentage of its current Max HP deducted is twice the previous percentage; the other surviving robots each have 5% of their current Max HP deducted. - In each round, when a robot accumulates 3 Yellow Card Warnings, it also receives a Red Card Warning. Two-side Yellow Card: The operator interfaces of all Operators are obscured for 2 s, and all robots have 5% of their Max HP deducted. This is not counted toward any single robot's accumulated Yellow Card count. Note: If Yellow Card Warnings are received consecutively, the operator-interface obscuring times are added together and the 30 s timer is reset. If a robot's remaining HP is less than or equal to the HP deducted by a penalty, its HP is reduced to 1. The chassis power-cut and operator-interface obscuring durations caused by violation penalties are independent of the match mechanisms and are not added to them. 
Red Card Warning (Ejection)	- Robot Ejection: - If a robot is Ejected before the 15-Second Referee System Self-Check Period, it may not take the field and must be removed from the Battlefield. No other robot may substitute for it in any remaining round of that Match. - If a robot is Ejected during the 15-Second Referee System Self-Check Period, the Red Card Warning is issued after the match starts. - If a robot is Ejected during a match, its HP immediately becomes 0, its video feed and multi-robot communication are disconnected, and it cannot undergo Revival. - Ejection of field team members: The Referee requires the Ejected person to leave the competition area immediately. No other field team member may replace them in any remaining round of that Match. The robot controlled by an Ejected Operator is also Ejected for the current round and may not take the field in any remaining round of that Match; no other robot may substitute for it.

Penalty type	Description
Loss by forfeit	If a Loss by forfeit penalty is issued for one round (hereinafter ‘round forfeit’), the detailed rules are as follows: 3V3 Match: - If the forfeit penalty is issued before the five-minute Round, including during the Setup Period and the 15-Second Referee System Self-Check Period, the violating side’s Control Points are 0, its remaining robots have full HP, and the other side’s Control Points and robot HP are full. The round time is regarded as 300 s. - If the forfeit penalty is issued during the five-minute Round, the round ends immediately. The violating side’s Control Points are 0, its remaining robots’ HP is that at the end of the round, and the other side’s Control Points and robot HP are the values at the end of the round. The round time is the actual elapsed match time when the forfeit is issued. - If the forfeit penalty is issued after the five-minute Round, the violating side’s Control Points are 0, its remaining robots’ HP is that at the end of the round, and the other side’s Control Points and robot HP are the values at the end of the round. The round time is the actual elapsed match time when the forfeit is issued. Infantry Match: - If the forfeit penalty is issued before the two-minute Round, including during the Setup Period and the 15-Second Referee System Self-Check Period, the violating side’s robot HP is 0 and the other side’s robot HP is full. The round time is regarded as 120 s. - If the forfeit penalty is issued during the two-minute Round, the round ends immediately and both sides’ robot HP is that at the end of the round. The round time is the actual elapsed match time when the forfeit is issued. - If the forfeit penalty is issued after the two-minute Round, both sides’ robot HP is that at the end of the round. The round time is the actual elapsed match time when the forfeit is issued. If a Loss by forfeit penalty is issued for one match (hereinafter ‘match forfeit’), the penalty applies to all rounds of that match, and HP settlement for each round follows the descriptions above.
Disqualification from awards	- Participants are disqualified from awards. • Teams are disqualified from awards.
Disqualification	- Participants are disqualified from the competition and from awards. • Teams are disqualified from the competition and from awards. The team's results for this season are retained as a reference for the advancement of other teams.

6.2 Detailed Penalty Rules

This section introduces the details of penalties for violations. The rules numbered R# clearly indicate the rules that participating teams, contestants and participating robots must follow.

6.2.1 Personnel

6.2.1.1 General Rules

R1 A participating team must meet the requirements of the RoboMaster 2027 University League Participant Manual.

Penalties for violations: up to disqualification of the offending party.

R2 participants and their actions must not interfere with the normal operation of official equipment, event processes, or the normal work of RMOC personnel.

Penalties for violations: up to disqualification of the offending party.

R3 A participating team must not set up its own wireless equipment or use walkie-talkies in competition-related areas, including but not limited to the preparation, inspection, waiting and competition areas.

Penalties for violations: up to disqualification of the offending party.

R4 Participating teams must not damage official equipment, including but not limited to equipment in the competition, waiting, preparation and inspection areas.

Violation penalty: The violating side may be disqualified from the competition and required to pay compensation at the item's full price.

R5 Except for Battlefield personnel who enter the waiting area and competition area due to competition needs, other participants are not allowed to enter the waiting area and competition area without special reasons.

Penalties for violations: up to disqualification of offenders.

R6 Field team members who entered the waiting or competition area for match purposes must not leave those areas or replace another field team member without the referee's permission.

Violation penalties: Violators are not allowed to enter the waiting area and competition area during the game, and the maximum penalty for violators is disqualification.

R7 Except for projectiles preloaded in the inspection area, participating teams must not bring their own match projectiles into the waiting or competition area.

Penalty for violation: projectile confiscated and verbal warning given. If the warning is invalid, the offender will be disqualified from the competition at most.

R8 After a Match, field team members must immediately power off the robots, carry them off the field and remove all projectiles from them in the projectile-clearing area.

Violation penalty: The violating robot is held in the projectile-clearing area until its projectiles have been removed.

R9 After a Match, field team members must return all projectiles used in the match in the projectile-clearing area.

Violation penalty: Confiscate the projectiles and bar the violating person from entering the field for subsequent matches in the current division; at most, disqualify that person from the competition.

R10 Except for emergencies, the participating team must arrive at the inspection area for pre-match inspection at a designated time before the start of each game (see "5.1 Pre-match Inspection" for details), and the team must arrive at the waiting area and be ready 10 minutes before the start of each game.

Violation penalty: up to a Loss by forfeit in the current match.

R11 Without the permission of the referee, the robot may not be powered on, debugged or repaired in non-Battlefield areas after the robot has passed the inspection and before the end of the game.

Violation penalty: up to a Loss by forfeit in the current match.

R12 The identity and number of personnel entering designated areas such as the preparation area, inspection area, waiting area, and competition area for each team must meet the requirements.

Penalties for violations: up to disqualification of the offending party.

One of the R13 Battlefield personnel must wear the "captain" armband, and the armband must not be obscured.

Penalties for violations: up to disqualification of offenders.

R14 Without the permission of the referee, Battlefield personnel entering the arena are not allowed to communicate with the outside world.

Penalty for violation: verbal warning. If the warning is invalid, the offender will be disqualified from the competition at most.

R15 Battlefield personnel are not allowed to use official equipment power supplies to power their own equipment in the competition area, but they can bring their own power supplies.

Penalty for violations: verbal warning. If the warning is invalid, the offender will be given a red card. The offending party may be disqualified from the competition.

R16 Except for special circumstances, Battlefield personnel are prohibited from wearing slippers into the arena.

Penalties for violations: up to a red card warning to the violator.

R17 During an Appeal, the appealing side must provide valid evidence or cite the relevant provisions in the governing documents on which it relies.

Penalties for violations: up to disqualification of offenders.

The R18 captain has management responsibilities and must ensure: during preparation and participation in the competition:

- Participating robots must pass the review of finished robots and open source robot usage specifications;
- Team Mentors, Reserve Members and other personnel who are ineligible for awards must not commit any violations while participating.
- Stop other violations promptly.

Violation penalty: The captain’s management responsibility is determined according to the violation; at most, the violating person is disqualified from the competition.

6.2.1.2 Battlefield Rules

R19 Battlefield personnel are required to wear goggles when entering Battlefield.

Penalties for violations: Offenders will be ejected from Battlefield.

During the R20 official technical suspension, Battlefield personnel cannot repair other faults except Referee System related modules.

Penalty for violation: verbal warning. If the warning is ineffective, a red card warning will be issued to the violator.

R21 At the end of the Setup Period, field personnel must return to the designated area outside the Battlefield. During the match, field personnel must not leave that area without the referee's permission.

Violation penalty: Verbal warning. If the warning is not valid, issue a red card warning to the offender.

R22 After the Setup Period ends, field team members must place the Sentry Robot's debugging Remote Controller in the designated area at the Battlefield entrance, and must not use it to debug the Sentry Robot after the Five-Second Countdown begins.

Violation penalty: Before the five-minute Round, issue a Verbal Warning; if it is ineffective, issue a Red Card Warning to the violating robot. During the five-minute Round,

issue a Red Card Warning to the violating robot.

R23 After the Five-Second Countdown starts, field personnel must not operate the Remote Controller for a robot on the field that is located outside the Operator Room.

Violation penalty: Issue a Red Card Warning to the violating robot; at most, the round is forfeited.

R24 Field personnel must ensure that their robots operate safely and do not harm any people or equipment in the competition area.

Penalties for violations: The violating party shall bear corresponding responsibilities.

6.2.1.3 Operator Room Rules

R25 In addition to the Operator corresponding to the playing robot, personnel from other venues are not allowed to enter the Operator Room.

Violation penalty: Verbal warning. If the warning is not valid, issue a red card warning to the offender.

R26 Without referee's permission, within the 15-second Referee System self-test phase and in Round, the Operator shall be located in the corresponding Operator Room to operate the control equipment of the corresponding robot, and shall not move the position after the end of the preparation phase.

Violation penalty: Verbal warning. If the warning is not valid, issue a red card warning to the offender.

R27 During a match, each Operator may use at most 1 Remote Controller and 1 Custom Terminal.

Violation penalty: Verbal warning. If the warning is not valid, issue a red card warning to the offender.

R28 Do not use your own headphones or computer in the Operator Room (except Custom Terminal).

Violation penalty: verbal warning; if the warning is ineffective, the violating side loses the round.

R29 Each Operator Room operating station provides at most 1 RJ45 network port, which may connect only to a Custom Terminal.

Violation penalty: Verbal warnings and prohibition on using the violating equipment. Depending on the impact caused by the equipment, the violating side may be disqualified from the match at most.

6.2.1.4 Robots

Robots and related equipment for R30 matches must be checked before the game.

Violation penalty: the violating side loses the round.

R31 In the first round of a match, the robot lineup must meet the minimum starting lineup requirement.

Violation penalty: Forfeit the current match.

R32 Robots must comply with the RoboMaster 2027 University Series Robot Building Specifications Manual.

Violation penalty: maximum disqualification of the violating party.



- RMOC will spot check the robot from time to time.
- Anyone reporting that a robot does not comply with the Robot Building Specifications Manual must provide corresponding evidence.
- During the competition, if the installation of the Speed Monitor Module does not meet the requirements, the module is deemed to be offline.

R33 While waiting in the waiting area, participants must not take a robot out of the waiting area without permission.

Violation penalty: Verbal warning. If the warning is invalid, issue a red card warning to the violator and the robot, up to disqualifying the violator.

R34 Robots must not exist or present safety hazards, including but not limited to short circuits, crashes, smoke, open flames, parts falling to the ground, gas cylinder explosions, etc.;

If a safety hazard exists or arises, participants must cooperate with the referees and carry out the corresponding actions.

Violation penalty: Before the field is cleared, field team members must resolve safety issues at the referees' request; otherwise, the violating robot may not take the field. If the issue arises during a match, issue a Verbal Warning; if it is ineffective, issue a Red Card Warning to the violating person or robot. If the safety hazard is serious, the Head Referee handles it according to '8 Abnormal Situations'.

R35 No robot may continuously launch projectiles outside of Battlefield.

Violation penalty: Verbal warning. If the warning is not valid, issue a red card warning for the offending robot.

R36 During the Setup Period and the 15-Second Referee System Self-Check Period, robots on the Battlefield must not leave their corresponding Starting Zones.

Violation penalties: Maximum red card warning for violating bots.

R37 During the Setup Period, any projectile that needs to be fired must be fired into the projectile-clearing bag provided by the referees.

Violation penalties: Maximum red card warning for violators and violating bots.

R38 During the race, the robot shall not be broken down into sub-robots or multiple subsystems connected by flexible cables, and shall not actively throw or launch its own parts.

Violation penalty: Issuing a red card warning to the violating robot.

R39 A subsystem or individual component resulting from disassembling a Heavy Robot must not leave the Workbench.

Violation penalties: Maximum red card warning for violating bots.

During the R40 competition, no surviving robot shall use its own structure to cover any Armor Module, nor shall it be deformed beyond the maximum stretch size.

Violation penalty: Issue a warning to the violating side based on subjective intent. If the concealment is deliberate, issue a Yellow Card Warning and a Verbal Warning when the violation occurs;

if the Verbal Warning is ineffective, issue a Red Card Warning. If the concealment is passive, issue one Yellow Card Warning when the violation occurs.

R41 In the preparation stage, the module or part replaced by the robot shall meet the requirements of "equivalent parts", see "5.3 Preparation stage" for details.

Violation penalties: Maximum red card warning for violating bots.

6.2.2 Interaction

6.2.2.1 Robot Interaction

R42 A robot on one side may not use any of its own structures to ram the other robot. If the dead robot blocks the critical movement path, it can be slowly pushed away.



In a collision between Ground Robots, the violating robot is the one the referee determines to be the active party.

Violation penalty: Issuing a warning to the violating robot based on the subjective intention and the degree of impact.

Table 6-3 Collision Violation Penalty Standards

Violation Level	Description
Yellow Card Warning	- Actively causing a head-on, high-speed collision • Actively pushing so that the opposing robot moves noticeably • Actively pushing the opposing robot to hinder its normal movement
Red Card Warning	- Actively causing a head-on, high-speed, repeated or violent collision • Actively pushing so that the opposing robot moves a considerable distance • Actively pushing the opposing robot to seriously hinder its normal movement

If a collision causes serious consequences for the opposing robot or affects match fairness, the violating robot (by robot number only) may not take the field in the next round of the same competition stage, and no other robot may substitute for it.

R43 A robot must not, through active interference, blocking, ramming or similar actions, cause any of its own structures to become rigidly connected to the opposing robot.

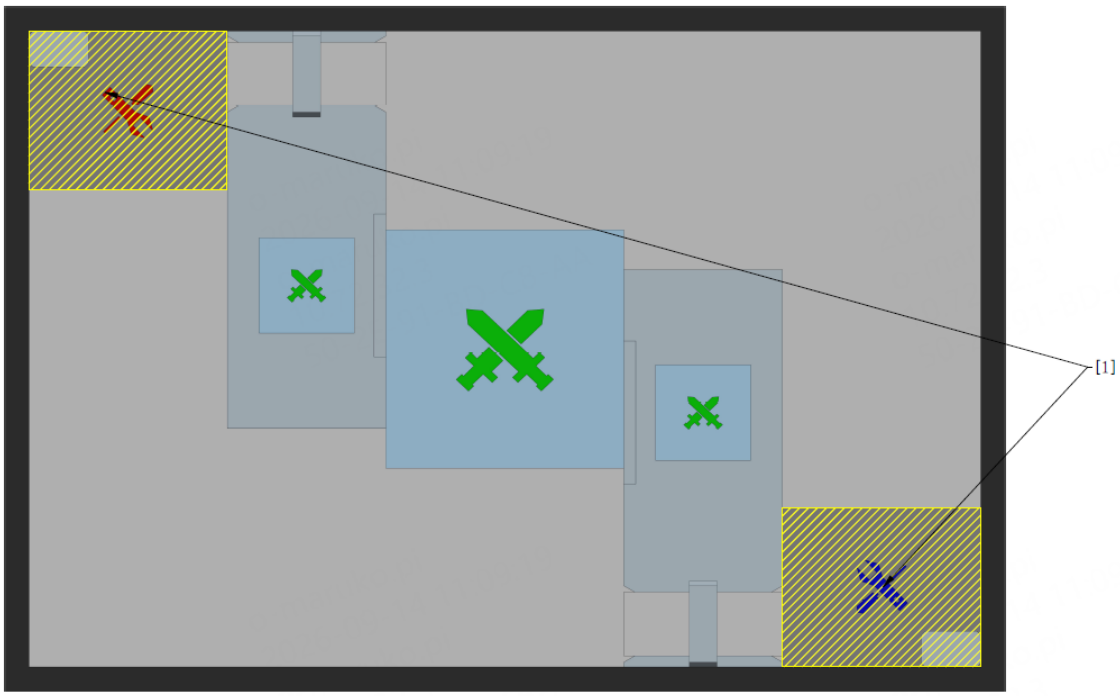
Violation penalty: Start timing when a Rigid Connection can be identified. Warn the violating robot according to the duration of the violation. When the violation exceeds 10 s, issue the first Yellow Card Warning, then issue one Yellow Card Warning every 20 s until the violating robot is ejected. Regardless of whether the robot is Alive, if the violation lasts more than 90 s, the violating side is given a Loss by forfeit for the round at most, depending on its intent.

R44 A robot must not use any contact-based means to interfere with the opposing robot’s HP Recovery.

Violation Penalty: Issue a yellow card warning to the violating robot.

6.2.2.2 Robot Interaction with Field Props

To ensure fair competition and that robots on the Battlefield can effectively obtain Buffs or supplies, supply Restricted Zones are established. One side's Supply Zone is a supply Restricted Zone for the opposing side's robots. Robots of one side or both sides, as applicable, must not enter a Restricted Zone, as shown in the figure.



[1] Supply
Restricted Zone

Figure 6-1 Schematic of Battlefield Restricted Zones

R45 A robot from one side shall not enter the supply restricted area.

Violation penalty: Issue a warning to the violating side according to the duration of the stay and the impact of the violation. When the violation exceeds 3 s, issue the first Yellow Card Warning, then one every 10 s until the robot is Defeated or abnormally offline. If staying in the Restricted Zone causes serious damage to an opposing robot, issue a Red Card Warning to the violating robot.

R46 Wireless debuggers and Movable Props used in the match must not be brought onto the field. During the match, robots may use only the official projectiles and Movable Props provided by the Organizing Committee on the field.

Violation penalty: The maximum disqualification of the violating party from the competition.

R47 During a match, robots must not damage Field Props or projectiles, or interfere with the normal operation of Field Props.

Violation penalty: The maximum disqualification of the violating party from the competition.

6.3 Serious Violations

If the following behaviors occur during the competition, it will be judged as a serious violation. For serious violations, RMOC will disqualify the offending party from the competition up to the maximum. If the act violates local laws and regulations, RMOC will cooperate with relevant departments to pursue the legal responsibility of the violator.

Table 6-4 Types of Serious Violations

Item	Type
1	Maliciously damaging the field, props or other official equipment, or the robots, equipment, etc. of other participating teams.
2	Falsification, impersonation or other dishonest participation behavior.
3	Modifying or damaging the Referee System, or using technical means to interfere with any detection function of the Referee System.
4	Situations that do not comply with the normative documents and are judged by the Chief Referee to be serious violations.
5	Other behaviors that hinder the competition, such as disobeying penalties, not cooperating with inspections, deliberate delay, disrupting order, abandoning or boycotting the match without cause.
6	Passive play, manipulating the match, and similar behaviors.
7	Giving, or illegally soliciting or accepting, property to or from others in order to obtain improper competition results or seek improper benefits.
8	Uncivilized or immoral words and conduct such as slander, abuse, improper gestures, malicious heckling and maliciously throwing objects.
9	Publishing, spreading or disseminating to the media untrue or irresponsible statements.
10	Deliberately attacking or ramming others, or committing acts that endanger one's own or others' safety.
11	Carrying dangerous or prohibited items.
12	Other behaviors that violate the spirit of the competition and are judged to be serious violations.
13	Other words and conduct that go against the core socialist values, violate sports ethics, violate public order and good morals, violate the competition style and discipline, cause adverse social impact, or violate laws and regulations.

7 Abnormal Situations

there will be a certain delay in referee's manual punishment and handling of abnormal situations. If it has a significant impact on the competition results, Chief Referee will determine the final handling result according to the actual situation.

If the following abnormal situations occur, they will be handled according to the corresponding methods, and the teams of both parties shall not have any objections. The handling methods are as follows:

- When serious safety hazards or abnormal conditions occur, such as battery deflagration, venue power failure, high-pressure gas cylinder explosion or on-site personnel conflict, etc., Head Referee will notify both Operators and penalize all robots through the Referee System. The results of the game will be invalidated and the game will restart after the hidden hazards or abnormalities are eliminated. RMOC will prioritize safety issues and will do its best to protect the robot to the extent possible, there is no guarantee that there will be no impact on the robot at all, and any loss or impact caused by safe disposal will be borne by the participating teams themselves.
- During the match, if non-critical props on the Battlefield are damaged—for example, if the floor adhesive is damaged—in a way that does not affect the fairness of the match, the match proceeds normally.
- During the match, if a robot's Light Indicator has an abnormal lighting effect, the match proceeds normally.
- During a match, if a key Battlefield prop has a logical or structural failure—for example, an abnormal network causes a robot to go offline or a Battlefield prop mechanism cannot operate normally—the Referee will handle the failure manually through the Referee System. If it cannot be handled manually, the Referee will notify both Operators and Eject all robots at the same time. The round ends immediately and its result is invalidated. The match restarts after the problem is resolved.
- During a match, if abnormal operation or structural damage of a key Battlefield prop affects fairness and the Head Referee does not promptly confirm this and end the match, so that the match continues when it should have ended and produces a winner and loser, the Chief Referee will verify the matter within 5 minutes after the match ends. The round result is then invalidated and the round must be replayed.
- If a serious violation clearly warrants a Loss by forfeit penalty but the Head Referee does not confirm and enforce it in time, the Chief Referee verifies it within 5 minutes after the match ends, voids that round's result, and additionally issues a Loss by forfeit penalty to the violating side.
- During the course of the game, if there are any situations that may affect the fairness of the game, within 5 minutes after the end of the game, Chief Referee will notify the captains of both sides of the situation and suspend the results confirmation process, and within 60 minutes thereafter, verify and inform the captains of both sides of the final treatment. Neither party can dispute the outcome.

8 Appeals

Each participating team has 1 Appeal opportunity for each competition site and event in RMUL 2027; opportunities cannot be accumulated. If an Appeal succeeds, the opportunity is retained; otherwise, 1 opportunity is consumed. When no Appeal opportunities remain, the Arbitration Committee will no longer accept Appeals from that team. After accepting an Appeal, the Arbitration Committee arbitrates the submitted materials and relevant evidence. The Chief Referee informs both sides of the result on behalf of the Arbitration Committee, which has final authority to interpret the Appeal result.

The following are not grounds for appeal:

- Verbal Warning, Yellow Card Warning and Red Card Warning in Violation Penalties.
- The type and process for initiating a technical timeout.
- The Referee System onboard module indicates ‘Routine Battle Damage’.

An Appeal may not be initiated more than 5 minutes after signing the result confirmation form or after a match ends.

8.1 Appeal Process

A participating team that wishes to file an Appeal must follow this procedure:

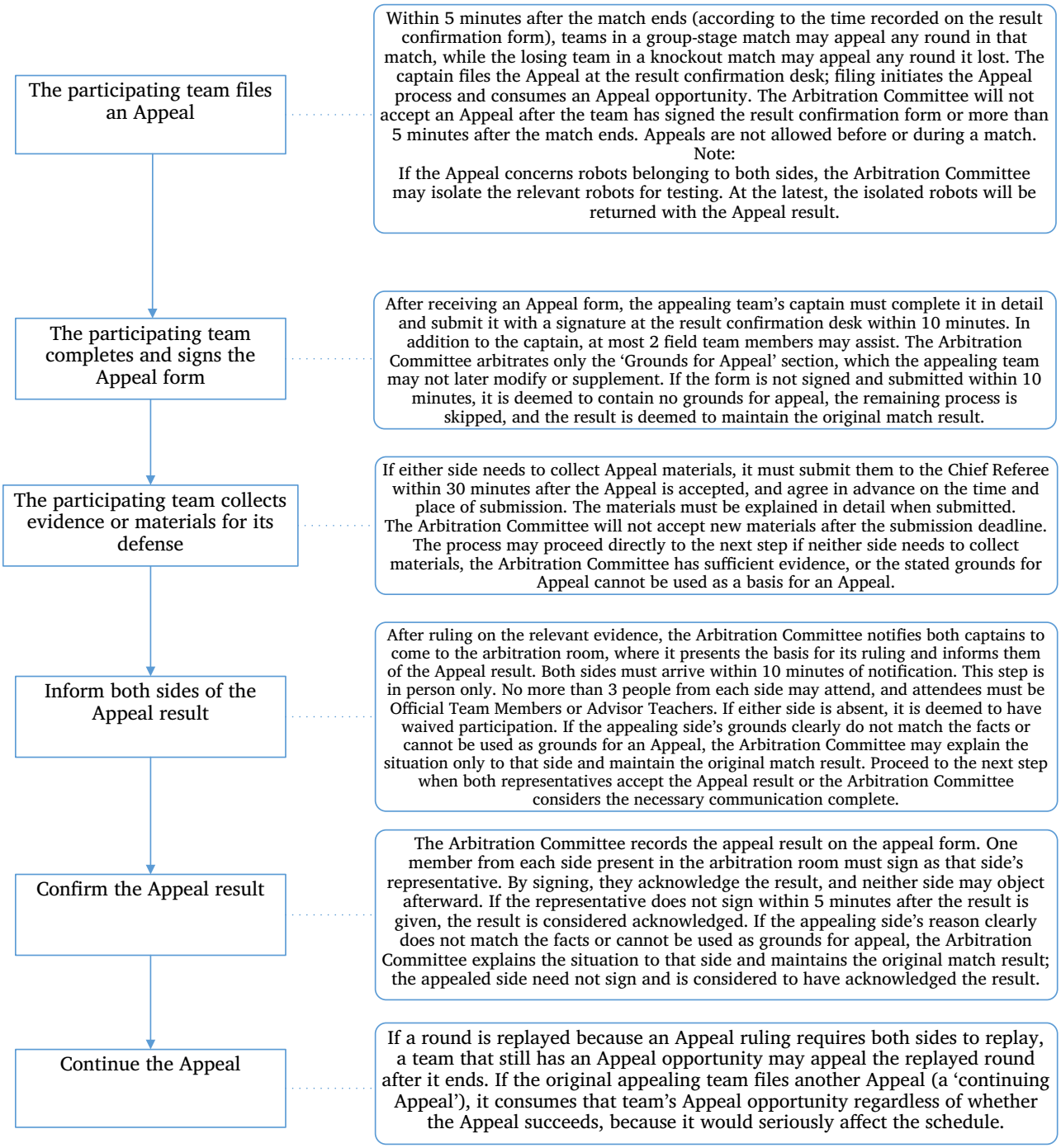


Figure 8-1 Appeal Flowchart

8.2 Appeal Materials

Materials submitted by a participating team for an Appeal must be no larger than 500 MB per file and may include no more than 10 files.

8.3 Appeal Results

Appeal results include maintaining the original match result, a Loss by forfeit for the respondent, or a replay by both sides. After the Arbitration Committee confirms the Appeal result, neither side may object.

- Successful Appeal: the respondent receives a Loss by forfeit, or both sides replay the match.
- Unsuccessful Appeal: the original match result is maintained.

If, during the Arbitration Committee's communication step, both sides are told to replay but neither accepts, the Appeal is unsuccessful and the original match result is maintained.

- Provided the overall schedule is not affected, a replay is generally scheduled after all matches that day have ended; the actual schedule governs.
- The replay follows the regular match procedure. Both sides must play at the time specified by the Organizing Committee and comply with the relevant rules.

If the Appeal result affects the win-loss result of the match, the match result must be determined again.

Example: The original BO3 match result is 1:2: Red won Round 1, and Blue won Rounds 2 and 3. After the match, Red appeals Round 2, and the Appeal result requires both sides to replay that round. If Red wins the replay, the final match result is 2:0; if Blue wins the replay, the final result is 1:2.

Appendix Pre-Match Procedure Description

Pre-Inspection

Pre-inspection is a check performed before formal Pre-Match Inspection to help teams find and resolve robot problems in advance. RMOC will do its best to help teams troubleshoot and handle robot faults during pre-inspection and adaptation training, but a pre-inspection result is not equivalent to the formal match inspection result. If a robot does not meet the inspection pass criteria during formal inspection, it may not compete.

Pre-Match Inspection Robot Carrying Restrictions:

- 3V3 Match: Each team may bring at most 4 robots for pre-inspection.
- Infantry Match: In each event, each team may bring at most 2 robots for pre-inspection.

Adaptation Training

Adaptation training is a debugging and familiarization opportunity for participating teams before the formal match. Its main purposes include:

- Familiarity with the match environment and procedure.
- Testing and adjusting robot status and strategy.

Adaptation training duration and opportunities:

- 3V3 Match: 15 minutes total, divided between 1 round and free debugging time.
- Infantry Match: 10 minutes total, divided between 1 round and free debugging time. A team participating in both the 3V3 Match and Infantry Match may attend adaptation training only for the 3V3 Match.

Adaptation training rules:

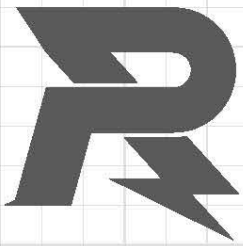
- The training schedule is compact and focused on debugging. By default, training is not paused unless major official equipment fails.
- If an onboard Referee System anomaly occurs, the RMOC may grant an official technical pause of up to 2 minutes to address the fault; this pause counts toward the total adaptive-training time.
- Team Technical Pauses are not available during adaptation training.

Rules for the free debugging stage:

- For projectile-firing tests, a participating team may ask the Referee to lend it a projectile-clearing bag and may fire only into that bag.
- The Round status of the free debugging stage is "not started", and the participating teams need to adapt themselves and be familiar with the switching and requirements of each stage of the competition.

Eligibility and Phase Segmentation:

- Robots that fail to pass the inspection on time or have safety hazards are not allowed to participate in adaptive training.
- Robots that do not pass the inspection due to serious non-compliance with the inspection criteria are only allowed to participate in the free debugging stage in adaptive training.
- The number of robots participating in adaptive training must not exceed the maximum robot lineup limit specified in each competition.



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